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A PROJECT REPORT ON

CHATPCCOE

SUBMITTED TO THE PIMPRI CHINCHWAD COLLEGE OF ENGINEERING
AN AUTONOMOUS INSTITUTE, PUNE
IN THE FULFILLMENT OF THE
REQUIREMENTS
FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF TECHNOLOGY
COMPUTER ENGINEERING
(REGIONAL LANGUAGE)

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2025-26

CERTIFICATE

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CHATPCCOE

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ABSTRACT

Now more than ever, our schools are trying to take advantage of technology to work, collaborate and learn in better ways. Most of the campuses in our world are still working with numerous disparate tools to facilitate communication between people through instant messages, emails and informal groups. This will surely result in the inability to get notified of announcements, receive the same information numerous times and make it hard to search for any information you may need; it makes collaboration and task management very challenging when things are so disjointed. Therefore, to solve this issue, ChatPCCOE has been built. This website allows people on campus to collaborate, featuring a multitude of capabilities such as a secure login, instant messaging functionality, a social feed and an organised notification system that make it incredibly easy for students, faculty and administrative staff to communicate with one another. ChatPCCOE makes use of technologies such as React, TypeScript and Supabase to provide for a fast and efficient web application, all while maintaining the safety of the user. The developers behind ChatPCCOE aimed to solve the issue of difficult communication within a campus and to provide a place where students and faculty can disseminate information, collaborate on group work, and accomplish their academic needs. ChatPCCOE is a practical tool for modern schools since it will allow anyone to communicate, collaborate, and learn with one another.

TABLE OF CONTENTS

LIST OF ABBREVATIONS

ABBREVIATION	ILLUSTRATION
API	Application Programming Interface
Auth	Authentication
BaaS	Backend-as-a-Service
CSS	Cascading Style Sheets
DB	Database
FK	Foreign Key
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
HMR	Hot Module Replacement
ICT	Information and Communication Technology
JWT	JSON Web Token
JS	JavaScript
LMS	Learning Management System
PK	Primary Key
PKG	Package
PostgreSQL	Post Relational Database Management System
PRN	Permanent Registration Number
RLS	Row-Level Security
RPC	Remote Procedure Call
React	React JavaScript Library
SDG	Sustainable Development Goal
SDLC	Software Development Life Cycle
SRS	Software Requirements Specification
TypeScript	Typed JavaScript Superset
UI	User Interface
URL	Uniform Resource Locator

LIST OF FIGURES

LIST OF TABLES

1. INTRODUCTION

1.1 OVERVIEW

Today schools are rapid using technology for betterment of both students and teachers. For making communication easier between the two and easy collaboration and learning is effective. The goal of this project is not to just make the daily usage of digital resources simpler but to form an integrated system beneficial to communication and interaction. It is possible to enhance the system by introducing new users and features when and where required. Many times these will cause miscommunications and can be confusing to monitor which is sent, causing students to not receive updates where applicable or teachers to be uncertain that students have been notified. For such issues there is a need for a single consolidated system. The system ChatPCCOE that we developed is a web application for Pimpri Chinchwad College of Engineering that enables the ease of communication between the two parties, aids access to required resources and provides the framework for teamwork. This website has functionality for login, user profile management, academic resource sharing system, announcements and event details etc.

For the implementation of this website the part handled by the student (the front-end) was implemented using the JavaScript library React.js with TypeScript as the programming language, with the presentationlayer built using Tailwind CSS which makes the interface responsive on different devices. On the back end, we have implemented login authentication, data base management etc using Supabase.

The ChatPCCOE represents a step towards making schools and institutes digital. It is designed to enhance the experience of the student and teachers by streamlining communication and collaboration. The project aims not to just simplify the everyday use of digital resources but to create an integrated platform that benefits communication and interaction between users and the entire system can be expanded with further users and features as per requirement.

1.2 MOTIVATION

Main reason why we are creating ChatPCCOE is because college doesn't have a system for people to talk to each other. Now all teachers and students are using so many tools like WhatsApp group chats, emails and other tools to send notes, Homeworks, important dates...

Which make it very difficult for everyone to keep up with it and confusion will appear. That information will be hard to find later and hard to access. ChatPCCOE is also made to give opportunity to students to help each other and share information, we don't have something like it now which give the opportunity to students to talk to each other, sharing information that have learn, keeping up to date with what is happening at the college. It also require a system which can be managed by that many users and data yet maintain correctness, using Supabase. So it enables you to know that people are actually who they say they are and that any data is actually true. We also wanted to use web technology, so we're using React.js, TypeScript for the interface, and for the backend we are using Supabase, since data is easy to manage, and make sure the user is who they want to be. Make's it faster to build and maintain. And it is the reason that we are using ChatPCCOE. This is why the answer of the university is: more lectures, talks and communications, etc. This is what it will do to our college and that is why we are using it

1.3 PROBLEM STATEMENT AND OBJECTIVES

1.3.1 Problem Statement

There is no platform in Pimpri Chinchwad College of Engineering to communicate and share information between students and professors. Currently, communication is performed using multiple medium's like IM's, Emails, unofficial groups, which makes the information scatter over the area and also very difficult to keep track of things. Due to this many people are missed about the announcement, the same resources gets posted more then one time, and are very difficult to retrieve in times when needed. It's

very hard for the students to work on some project together. The college does not provide a system where there can be a surety that the information available is accurate and delivered to each person on the right time. Also the present system in use is not secure enough; anybody can have access to the available information which might cause harm if the wrong person will gain access to it and misuse. Also the system in use is not enough to handle multiple number of users and sufficient amount of academic data. Hence all these factors considered above, Pimpri Chinchwad College of Engineering needed a portal which is secure, can be managed by the current college needs in terms of data both about the users and academic matter and makes communication and interaction easier between student and teachers.

1.3.2 Objectives

ChatPCCOE aims to build a communicative, collaborative resource-sharing platform for users to receive academic information in one single space. The platform uses secure logins and access permissions to ensure appropriate access to information. Additionally, it will allow users to share and locate such items as announcements, notes and event information which will eliminate the need for multiple platforms.

ChatPCCOE's second goal is to improve communication between faculty and students, including the sharing of knowledge and resources in an organized manner. The architecture of ChatPCCOE is designed to effectively support the large number of users, transactions and data. The Supabase platform will be used for all backend services allowing for the safe and timely storage of information and consistent performance from the entire solution.

Ultimately, ChatPCCOE plans to build a simple-to-use, user-friendly interface with a range of different technologies that ensure high usage levels while delivering good end- user experiences. In summary, ChatPCCOE wants to create an efficient scalable system that enhances academic functionality and enhances collaboration and communication within the larger academic community.

1.4 SCOPE OF THE WORK

The ChatPCCCE project aims at building a platform where students of PCCCE can easily interact with each other and work collaboratively as well as sharing resources. The targeted users are students and faculty of PCCCE, as they are also able to obtain academic information, generate ideas and learn about all activities and events happening at the PCCCE campus. The main features of the proposed website will consist of, registration and sign-in features, user profile management, school assignment submissions, and notifications. All of the aforementioned features will be integrated into one interface, making it easy to use.

In addition to developing this type of user outcome, the project will also create an organized database using Supabase to facilitate rapid access to the data required by the users while maintaining current data for all users. The system will be designed for a large number of users while still being functional and by providing sufficient processing power to accommodate all of the users. Secure storage of data and ease of access to the data are critical to ensuring that the institution can utilise the system effectively. The final outcome of this project is a functional website; the original intended use for the proposed website was to support computer-based applications, but the site is currently only available to computer use and not mobile devices (e.g. cellphones) as initially planned.

Currently, there are additional features (ie, recommendations through artificial intelligence) that will not be implemented immediately due to uncertainty on the availability of data from the school district and multiple databases containing large amounts of information. These additional features may be available at a later date. The immediate goal is to provide an easy-to-use website to facilitate communication and collaboration among users, primarily based on the premise of cooperation (ChatPCCOE).

1.5 METHODOLOGIES OF PROBLEM SOLVING

The ChatPCCOE development process centres on an organised Software Lifecycle Development (SLCD) approach. The first step to developing ChatPCCOE is performing a requirements analysis based on a review of what people's needs and problems are through observation and research, in order to determine how our proposed solution meets those needs or issues.

The next part of the ChatPCCOE development process is the creation of a system proposal build out the system design. The system design is where we will create a blueprint for how the system will be built and developed. We will develop an architectural design for the system, and then the database schema design, and finally the user interface (UI) for the system. One of our requirements is for the system to accommodate multiple user types and handle significant amounts of data and operate in a secure and reliable manner.

After the system design has been completed, the next part of the ChatPCCOE development process is building the system. The ChatPCCOE front-end (UI) is built

using React.js in conjunction with TypeScript, which provides an aesthetically pleasing UI that performs optimally. The ChatPCCOE back-end is built using Supabase, and uses Supabase to assist with providing key functions such as authentication, database management and overall system maintenance / upgrades.

The final phase of the ChatPCCOE development process is to conduct system testing to ensure the system functions as intended/specified.

2. LITERATURE SURVEY

2.1. REVIEW OF RECENT LITERATURE

There has been a significant increase in the growth of web-based platforms over the last few years. These platforms focus on managing academic resources and communication, as well as creating opportunities for collaborative work among schools and colleges. The recent surge of web-based technologies has resulted in an increase in the use of Learning Management Systems (LMS), student portals, and communication applications as a means of providing easy access to academic resources and enhancing communication capabilities. These systems assist schools and colleges with activities such as course management, assignment tracking, announcements, and verification of user identity. The advent of frontend frameworks, such as React.js and user interface (UI) libraries (e.g., Tailwind CSS), has enabled LMS to become more interactive and user- friendly. Furthermore, the use of backend-as-a-service (BaaS) solutions, such as Supabase and Firebase, has simplified backend development. BaaS solutions provide authentication, database management, and real-time data synchronization.

Numerous pieces of research have concluded that web applications should be significant because scalability, performance, and security are paramount to the offered service by the web applications; research has used Next.js and Supably and cloud architecture applications to illustrate how developers are migrating away from the use of back-end systems towards the use of more efficient and scalable applications. In addition, the application of Next.js, Supably, and cloud architecture allows developers to rapidly create applications without the use of physical server infrastructures, reducing both the time and cost associated with creating applications. Additionally, the research has illustrated both student management systems and collaborative platforms as having available technologies; however, these technologies do not provide an integrated platform for communication and resource sharing. This shows that there is not only a growing need, but also a growing trend towards the merging of multiple functionalities into a single integrated solution.

Even with the many technological advancements, a lot of systems now in place still have issues, such as poorly organized communication, little to no personalization, and trouble with how data is managed. Most platforms today are either too generic or do not meet the specific needs of an organization, which reduces usability and engagement by the users. Research shows there is a clear need for a secure, user-focused platform that will combine communication, resource sharing, and administering academics. Our goal at ChatPCCOE is to provide this type of solution based on current technologies (React.js, TypeScript, Tailwind CSS, and Supabase) and to offer a single location-based solution for Learning Management Systems and student management – and we believe our platform will help to provide a better overall experience by addressing all the needs of the organization.

2.1.1 Literature Survey

Table 2.1 Literature Survey

Title	Objective	Technology	Published Date
Modern Learning Management System using React and Supabase	To develop a scalable LMS using React and Supabase for real-time academic interaction	React.js, Supabase, PostgreSQL	March 2024
Next.js Based Student Collaboration Platform	To create a collaborative platform for students using serverless architecture	Next.js, Supabase, Tailwind CSS	January 2024
Cloud-Based Student Management System using BaaS	To design a cloud-based system using Backend-as-a-Service for efficient data handling	Firebase, Supabase, React.js	December 2023
Real-Time Academic Portal using Supabase	To implement a real-time academic portal with authentication and database services	Supabase, React.js, TypeScript	August 2023
Web-Based Communication Platform for Educational Institutions	To develop a centralized communication system for students and faculty	React.js, WebSockets, Tailwind CSS	July 2023
Secure Authentication System using Supabase and Auth0	To enhance security in web applications using modern authentication services	Supabase, Auth0, Next.js	February 2024
Responsive College Portal using Tailwind CSS	To design a responsive and user-friendly college portal interface	Tailwind CSS, HTML, JavaScript	May 2023

Serverless Web Application Architecture for Education Systems	To implement serverless architecture for scalable educational platforms	Supabase, AWS, Next.js	September 2023
Comparative Study of Firebase and Supabase for Web Applications	To analyze performance and scalability of Firebase vs Supabase	Firebase, Supabase	April 2024
Integrated Academic Resource Sharing Platform	To develop a platform for efficient sharing of academic materials	React.js, Supabase, Cloud Storage	June 2024

2.2 MOTIVATION

Based on the analysis of existing systems and the information given about what is available, it is clear that significant advances have been made in creating space for education and using systems on a web-based level for schools. Many of these systems perform one specific function, such as facilitating student learning or managing student files, or providing a communication medium for users; however, they often exist independently, which results in poor user experience and thus inefficiency. This leads to students and teachers needing to switch across multiple platforms to access varying degrees of information, causing confusion. Additionally, these systems require complex backend programs that add difficulty in developing and maintaining them..

Another major issue with the systems is their inability to allow individuals to customize their experience and communicate synchronously with one another. Some systems, such as Facebook, provide users with the ability to communicate; however, they do not provide a collaborative environment for users to share ideas collaboratively or a place to actually complete tasks together. Additionally, systems that were built to accommodate fewer users face several performance and scalability issues. Security and privacy of data are other issues for systems that do not have reliable user authentication and authorization processes. While newer technologies such as Supabase and Serverless will address many of the above issues, adoption of these new technologies in education is still relatively low.

From what we've discovered we know we want a centralized system that can accommodate a large number of users and is secure. Additionally, this system will help group different items together into one location (ChatPCCOE does this by providing users with a single location to communicate, share resources and manage their things). The use of technologies such as React.js, TypeScript, Tailwind CSS, and Supabase allows the user to interact with others in real time, manage data properly, and improve the overall user experience. Therefore, ChatPCCOE provides a better solution compared to current systems available. With these new technologies combined to provide a complete and practical solution; ChatPCCOE's ability to group multiple items together into one place makes it a great fit for all types of users!

3. SOFTWARE REQUIREMENTS SPECIFICATION

3.1. FUNCTIONAL REQUIREMENTS

The users of the ChatPCCOE system will be able to utilize certain features and functions that will allow them to interact with the system and perform essential functions in order to be able to complete their job functions. The system has been designed for different users such as students and administrators, and the system will ensure that users are able to access and view only the appropriate information.

Users are able to log into ChatPCCOE using their school credentials. After a user logs into the system, he or she is able to access his or her dashboard, which contains important information such as announcements, share items and other recent activity. Users are also able to create and maintain their profiles within the ChatPCCOE system, where they can update their own information, photographs, and school-related information. As a result, users within the ChatPCCOE system will have an opportunity to get to know one another and communicate with one another easily

Users have the ability to share with one another through the ChatPCCOE system by uploading, accessing and downloading school work such as class notes and homework The ChatPCCOE system maintains everything in an easy to use manner, so it will be easy for users to find what they need to complete their tasks There will also be the ability for the administrators of ChatPCCOE to make announcements of school-related events or other on-campus activities by posting them as announcements which is an effective way to ensure that users are kept informed of the latest happenings at school

The ChatPCCOE system allows users to communicate with each other about their learning and to collaborate on projects with features such as postings and replies This collaboration allows users to share their knowledge and work, with others

Users of the ChatPCCOE system will also receive notifications when items have been updated such as when a new comment or resource has been posted Similarly, when The ChatPCCOE system sends a notification through WxCloud this ensures that all users will receive all of their notifications in a timely manner

In summary, the ChatPCCOE system is a communication and collaboration tool that helps users connect with one another, work together on shared learning and share schoolwork The ChatPCCOE system does this to assist users.

3.1.1 System Feature 1: User Authentication and Authorisation

platform. Individuals may register for the platform using their respective school or work emails and then later log in using a password that meets strict (but reasonable) criteria. The system utilizes Supabase to track all users who are currently logged in and to keep user passwords confidential. It also has a number of different levels of access designed to provide different capabilities to students than to those of administrators.

The system is built to safeguard data and prevent unauthorized persons from being granted access to it. The system also provides users with the ability to reset their passwords if they forget them, as well as manage their session time while logged in. Since the system relies on Supabase technology, a relatively small amount of code is required to implement the security features, making it easier to develop and maintain.

3.1.2. System Feature 2: Profile Management

Users can create and manage their own profile in the Profile Management function. Profiles can be updated at any time, including their name, branch, year and profile photo. This way, members will know who is who, and it will make it easier for members to interact throughout the system.

The Profile Management system performs very well to protect the profile information in their database, allowing for immediate retrieval of information when it is needed, as well as allowing members to view other members' profiles to aid members in meeting and communicating with one another. The Profile Management system has built-in control checks and validations of all profile information, ensuring consistent and accurate data throughout the system. Therefore, the Profile Management function in The system is key to assisting in the management of member profiles and is an effective system for establishing and maintaining member profile

3.1.3. System Feature 3: Resource Sharing System

The Resource Sharing Module is a very useful tool, as it allows users to upload, manage, and have access to important items such as notes, assignments, and other study materials. Resource Sharing is an effective tool for users because they are able to upload files. The files are then organized and categorized by subject and/or topic so that users can easily locate the resources they are searching for. Through the Resource Sharing Module, users can search for their resources by using keywords or various filters.

The Resource Sharing Module is also highly effective because it allows students to share knowledge with one another in a easy way, without having to depend on any Social Platforms. The Resource Sharing Module is completely integrated into the Supabase Storage service, as the resources or files uploaded are all stored in a scalable manner. The Resource Sharing Module also provides a means of Access Control to enforce who can upload or modify the files or resources available through the Resource Sharing Module.

3.1.4. System Feature 4: Announcement System

An announcement system is an opportunity for administrators to provide the community with updates about schoolwork, upcoming events, and daily happenings on campus. All announcements will be available to users via the user dashboard and will update in real-time as new announcements are posted.

This is important for ensuring that all users receive the information they need when they need it; thus, minimizing the likelihood of users missing an announcement or having to check multiple locations for an announcement. Supabase, a real-time data management tool, is used by the announcement system. Therefore, all users will receive notifications as soon as an announcement is made through the announcement system using Supabase for real-time updates.

3.1.5. System Feature 5: Notification System

The notification service informs users when announcements, uploads, or other activity occurs. This service creates and delivers notification messages without any user input, while also allowing all platform users to see these messages when they log into the system.

Supabase ensures all users receive their notifications away so that they remain engaged in the platform and are aware of the latest activity without requiring much work on their part. The notification system enables the user to keep track of recent notifications and activities, including new announcements and the uploading of new resources.

3.1.6. Functional Requirement

Table 3.1 Functional Requirement Summary

Feature	Description
User Authentication	Secure login and registration using Supabase
Profile Management	User can manage personal details
Resource Sharing	Upload and download academic materials
Announcements	Admin can post important updates
Notifications	Real-time updates for users
Interaction	Users can engage via posts/comments

3.2. EXTERNAL INTERFACE REQUIREMENTS

An integral part of the system design process is creating external interface requirements, which ensure that all components of the system can seamlessly interact with one another to provide a consistent user experience when using the system. It does this by clearly defining how the system works with other systems (i.e., computers) as well as how it connects to various hardware and software components along with communication networks.

3.2.1. User Interfaces

The ChatPCCOE is a simple, easy-to-use interface that functions across all platforms due to being built using both React.js and Tailwind CSS (both reactjs and tailwindcss work across all devices). There are many elements that exist in the ChatPCCOE user interface: dashboards, navigation menus, forms, and displays for content.

The ChatPCCOE user interface designers put a lot of thought into providing an easy user experience by choosing their color palette, typography, and layout style to increase the user's experience whilst using ChatPCCOE, regardless of platform. They also intended to create an effective interface across PC/laptop, tablet & smartphone.

3.2.2. Hardware Interfaces

ChatPCCOE does not need any computers or machines to work. You can use it on the computers you already have at home or work like desktops, laptops, tablets and smartphones. To use ChatPCCOE you need a keyboard and mouse. You can just touch the screen. The people who made ChatPCCOE wanted it to be easy to use on any device. So ChatPCCOE works well on older computers or devices that are not very powerful. This means anyone can use ChatPCCOE without having to buy devices or special equipment to make it work

3.2.3. Software Interfaces

The system works with Supabase. Supabase is what gives the system things like authentication and database management and real-time data synchronization.The frontend of the system talks to Supabase using APIs. This is how the system can get data and store data and make updates to the data.The system also works on web browsers like Google Chrome and Mozilla Firefox and Microsoft Edge. These web browsers are what let the system run and show the user interface.Using web technologies means the system works well on different platforms and it is compatible, with them.

3.2.4. Communication Interfaces

The way that ChatPCCOE talks to the server is really important. ChatPCCOE uses the internet in a way like when you use a web browser it uses HTTP and HTTPS to keep things safe. When you use HTTPS it means that what you send to ChatPCCOE and what ChatPCCOE sends to you is locked and hidden from people who should not see it.ChatPCCOE needs the internet to work properly so it can get information from the server and show it to you. ChatPCCOE also has a feature that lets it talk to you in real time and this is done with the help of Supabase. This means that when something changes you see it away and everyone who uses ChatPCCOE sees the same thing at the same time.

3.3. NON-FUNCTIONAL REQUIREMENTS

The system has to be good at what it does so we have -functional requirements. These non-functional requirements are about the quality of the system things like how fast it's how safe it is, how well it works and how well it handles a lot of users. The system needs to do things like performance and security and reliability and scalability. This means the system has to work well and do what the users want it to do so the users are happy, with the system.

3.3.1. Performance Requirements

The system is made to give people answers and handle data quickly. When you are on the website the pages should come up away and things like logging in getting data and uploading files should happen very fast. We use Supabase so the database can handle requests, in the way possible and deal with data in real time.

The website can handle users at the same time without slowing down too much. The people who built the website made sure to write the code in a way and optimize how the frontend works, which helps the website perform better.

3.3.2. Safety/Security Requirements

Security is an important part of ChatPCCOE. The system has security measures in place thanks to Supabase. This includes storing passwords in a way and managing user sessions. The system also makes sure that users can only use the features they are allowed to use.

Data that is sent over the internet is protected with HTTPS encryption. The system also checks everything carefully. Handles errors properly to stop bad things from happening to the security. This means that people cannot get to things they should not have access to and that data is not changed in a way. ChatPCCOE security is important. The system uses security measures, like encrypted password storage and session management to keep everything safe.

3.3.3. Non-Functional Requirements

Table 3.2 Non-Functional Requirement Summary

Requirement	Description
Performance	Fast loading and real-time updates
Security	Authentication, HTTPS encryption
Scalability	Can handle large number of users
Reliability	Minimal downtime
Usability	Easy-to-use interface

3.4. SYSTEM REQUIREMENTS

System requirements are the things that the platform needs to work. The platform needs to meet these system requirements so it can be developed and used. System requirements are, like a list of things that're necessary for the platform to run smoothly.

3.4.1. Database Requirements

The system uses Supabase, which is based on PostgreSQL for managing the data. This is where we keep all the information about users and resources and announcements and other important things.It is very good, at updating things away and finding the information we need quickly.Proper database design is really important because it helps keep the data consistent and accurate and makes sure the system can handle a lot of information. The system can handle a lot of data. Still work well.

3.4.2 Software Requirements (Platform Choice)

Table 3.3 Software Requirement

Component	Technology	Version
Frontend Framework	React	18.3
Language	TypeScript	5
Build Tool	Vite	5
CSS Framework	Tailwind CSS	v3
UI Components	shadcn/ui	Latest
State Management	Zustand	Latest
Animation	Framer Motion	Latest
Routing	React Router DOM	Latest
Backend (BaaS)	Supabase	Latest
Database	PostgreSQL	15+
Edge Functions	Deno Runtime	Latest
Package Manager	Bun	Latest

3.4.3. Hardware Requirements

To operate this system properly, you will require a minimum of 8gb of RAM, an acceptable processor, and internet connectivity to be able to use this system. The System is compatible to work with either Windows, Mac, or Linux operating systems. If you have a computer that meets the minimum hardware requirements stated above, you are able to use this system on whichever of the three operating systems is-

appropriate to you. In addition, this system will work with all versions of the three operating systems.

Table 3.4 Hardware Requirements

Parameter	Development	End Users
Processor	Intel Core i5 or equivalent	Any modern device
RAM	8 GB (min), 16 GB (recommended)	2 GB+
Storage	256 GB SSD (min)	N/A (web-based)
Internet	Broadband connection	1 Mbps or higher
Screen Resolution	1920 × 1080	320 × 568 (min)
Browser	Chrome 90+, Firefox 88+	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+

3.5. SDLC MODEL TO BE APPLIED

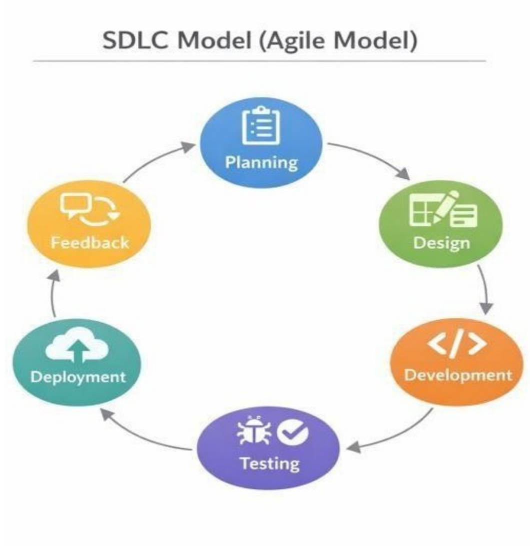

Figure 3.1 SDLC Model Diagram

ChatPCCOE uses the Agile Software Development Life Cycle (SDLC) methodology. This methodology has become quite popular within software engineering because of its adaptability and ease of modification. This differs from traditional methodologies such as the Waterfall Model, since Agile revolves around developing the system in smaller components called either iterations or sprints; thus allowing for improved versioning of the product based on user and other stakeholders feedback over time.

The Project is broken down into iterations. Each iteration's design and functionality focuses on developing proportionate levels of the overall system. For example, Initial iterations typically include developing the user interface and implementing user authentication using Supabase. Subsequent iterations will include adding profile management functionality, resource sharing and notifications systems. All developed functionality from an iteration will be validated after each iteration to verify it meets user requirements as specified. Feedback will then be used to make improvements for the following iteration. This provides multiple opportunities to eliminate errors early and to continually improve the system through each iteration of development.

With the Agile methodology, team members also perform well at working together; it makes communication more efficient, leads to faster decisions and promotes better problem-solving by allowing issues to be identified and resolved quickly in cycles of software development, without having to restart the entire project due to work comprised of more than one cycle, and has the ability to accommodate changes in the product requirements based on customer needs or requests for new features or improvements to the platform. As a result, The final product is more reliable if the requirements have been changing during the course of development allowing the system to be efficiently scalable and user-centered.

Agile SDLC provides a framework that has been used to build ChatPCCOE using a flexible method enabling the system to continuously develop into a high-quality solution that addresses changing business needs.

ChatPCCOE was designed with Agile principles because some interactive features such as authentication, resource-sharing and announcing real-time notifications, need to be iteratively developed over time due to the regular changes in requirements.

Therefore, ChatPCCOE will benefit from continual improvement through the use of the Agile methodology and is more likely to achieve a more effective and efficient system.

Overall the development of ChatPCCOE using Agile SDLC is an approach. It helps create a system that's of high quality reliable and meets the needs of its users.

4. SYSTEM DESIGN

4.1. PROPOSED SYSTEM ARCHITECTURE

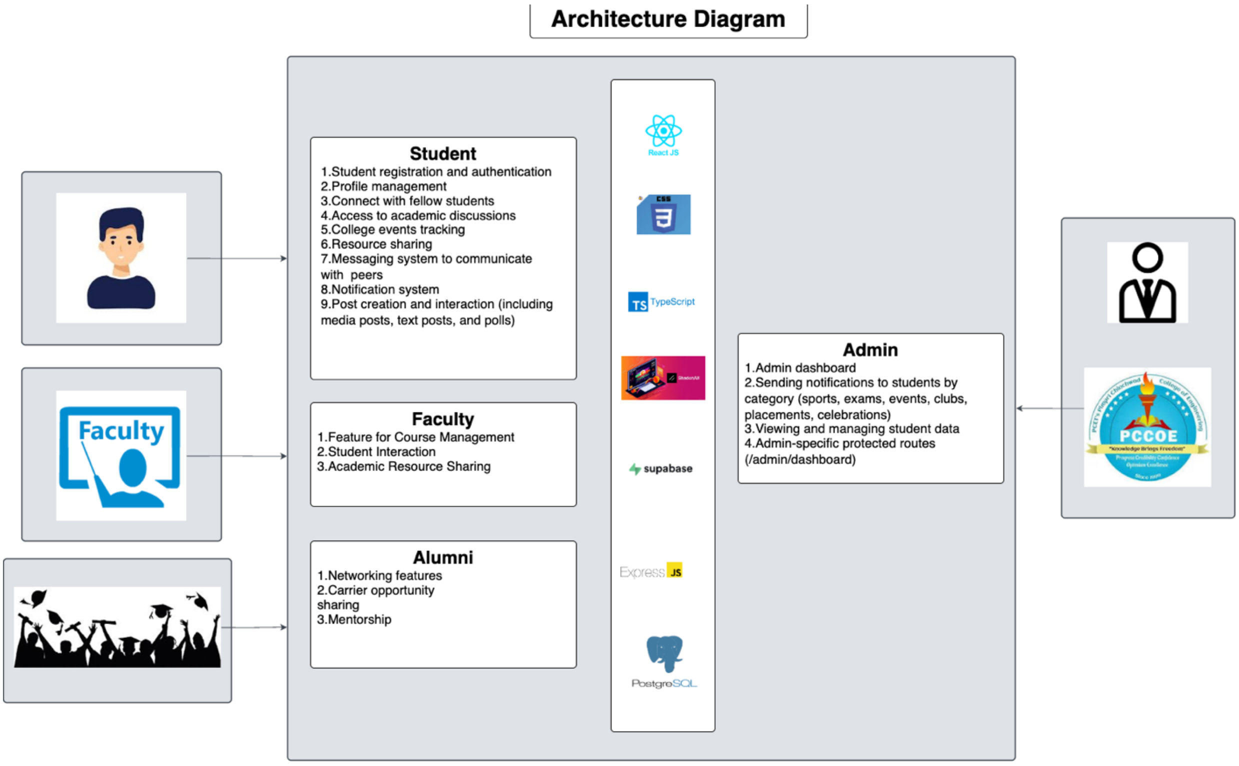

Figure 4.1 System Architecture

The ChatPCCOE system is made up of a simple and easy to use web application that combines different user roles, frontend technologies and backend services into one platform.The system has three parts: the user part, the application part and the backend service part. This way of doing things makes sure that each part does its job and it helps the system work better and handle data more efficiently.

At the user part the system supports types of users including Students, Faculty, Alumni and Admin. Each type of user interacts with the system in their way based on what they need to do. Students can do things like register and log in manage their profile connect with students look at academic discussions view events, share resources send messages get notifications and interact with posts and discussions. Faculty members can use features like course management, student interaction and sharing resources. Alumni users can use the system to network get mentorship and look for career opportunities. The Admin is in charge of managing the system.

The application part is built using frontend technologies like React.js, TypeScript and Tailwind CSS. React.js helps create an easy to use interface while TypeScript makes sure the code is safe and easy to maintain. Tailwind CSS is used to make the interface look good and work well on devices. This part of the system handles user interactions shows the user interface and talks to the backend services using API calls. It acts like a bridge between the users and the backend services making sure everything works smoothly and in time.

The backend service part is powered by Supabase, which provides a lot of services. Supabase handles things like authentication, database management, real-time data synchronization and storage. The database used is PostgreSQL, which's reliable and helps store data in a structured way. Supabase eliminates the need for a backend server, which makes development easier and improves scalability. It also simulates API behavior without needing a backend framework.

When a user does something on the system like logging in or uploading a resource the request goes from the React frontend to Supabase. Supabase then processes the request does what it needs to do and sends the response back to the frontend. The frontend then updates the user interface. This process makes sure everything works in time and it provides a seamless user experience.

The system also prioritizes security and performance. All communication between the frontend and backend is secure using HTTPS protocols to encrypt and protect data. The system also has role-based access control, which restricts access to features especially in the admin module. The system is designed to handle users at the same time making it scalable for use in institutions. Overall the ChatPCCOE system provides a flexible and efficient solution for managing academic communication and collaboration within the college.

The ChatPCCOE system is designed to be simple and easy to use. It combines many different parts into one platform. The system has three parts and each part does its own

job. The system supports types of users including Students, Faculty, Alumni and Admin. Each type of user interacts with the ChatPCCOE system in their way based on what they need to do. The ChatPCCOE system is built using frontend technologies and it handles user interactions shows the user interface and talks, to the backend services. The backend service part is powered by Supabase, which provides useful services. The ChatPCCOE system prioritizes security and performance. It is designed to handle many users at the same time.

4.2. DATASET/DATABASE DESIGN

Table 4.1 Database Design Table

Domain	Table Name	Description
User	profiles	Core user identity (id, full_name, email, role, avatar_url, status)
User	student_profiles	Extended student data (prn, branch, year, bio, interests)
User	admin_profiles	Extended admin data (employee_id, department, designation)
Social	social_posts	User posts (content, file_url, file_type, poll_id)
Social	post_likes	Like records with unique constraint on (post_id, user_id)
Social	post_comments	Comment records (post_id, user_id, content)
Social	polls / poll_votes	Poll definitions and vote records
Messaging	conversations	Conversation metadata (is_group, group_name)
Messaging	messages	Message records (content, message_type enum, file_url)
Messaging	group_members	Group role tracking (role: lead/admin/member)
Connection	connections_v2	Connection records with status enum (pending/accepted/rejected)
Notification	notifications	Broadcast notifications (title, content, category)

4.3. MATHEMATICAL MODEL

The ChatPCCOE system is made up of things that are put into it things that happen to those things and things that come out of it. This helps us see how information moves through the system and how all the parts work together. The system works in a way that we can expect. It does what it is supposed to do.

Let us say the system is like this:

S = {I, P O} Where:

I = The things that are put into the system like user names and passwords files that are uploaded and messages that are sent

P = The things that happen to the information like checking if the user's real, storing and getting data and sending notifications

O = The things that come out of the system like what you see on the page messages that are sent to you and getting to use files

The information that is put into the system goes through many steps like checking if it is correct making sure the user is real and working with the database. These steps change the information into something that makes sense and is shown to the user. The system makes sure that all the information that is put into it is checked before it is used so that the information stays good.

The way we think about the ChatPCCOE system also considers things, like keeping information safe and working well. For example only users who are real can use some

features. Getting data is made to be fast. This way of thinking helps us make a system that works well and is reliable. The ChatPCCOE system is designed to be good and to work efficiently.

4.4. ENTITY RELATIONSHIP DIAGRAM

The ChatPCCOE system is built around profiles. These profiles are for all the people who use the system. Each person can have a profile, like a student_profile or an admin_profile if they want to. People can make lots of social_posts get notifications and talk to each other.

When someone makes a social_post other people can like it comment, on it or even vote on a poll if there is one. Sometimes lots of people talk to each other so we use tables to keep track of who is talking to who. We also use these tables to keep track of who's connected to who.

The whole system is set up to show how people interact with each other talk to each other and use the ChatPCCOE platform to be social.

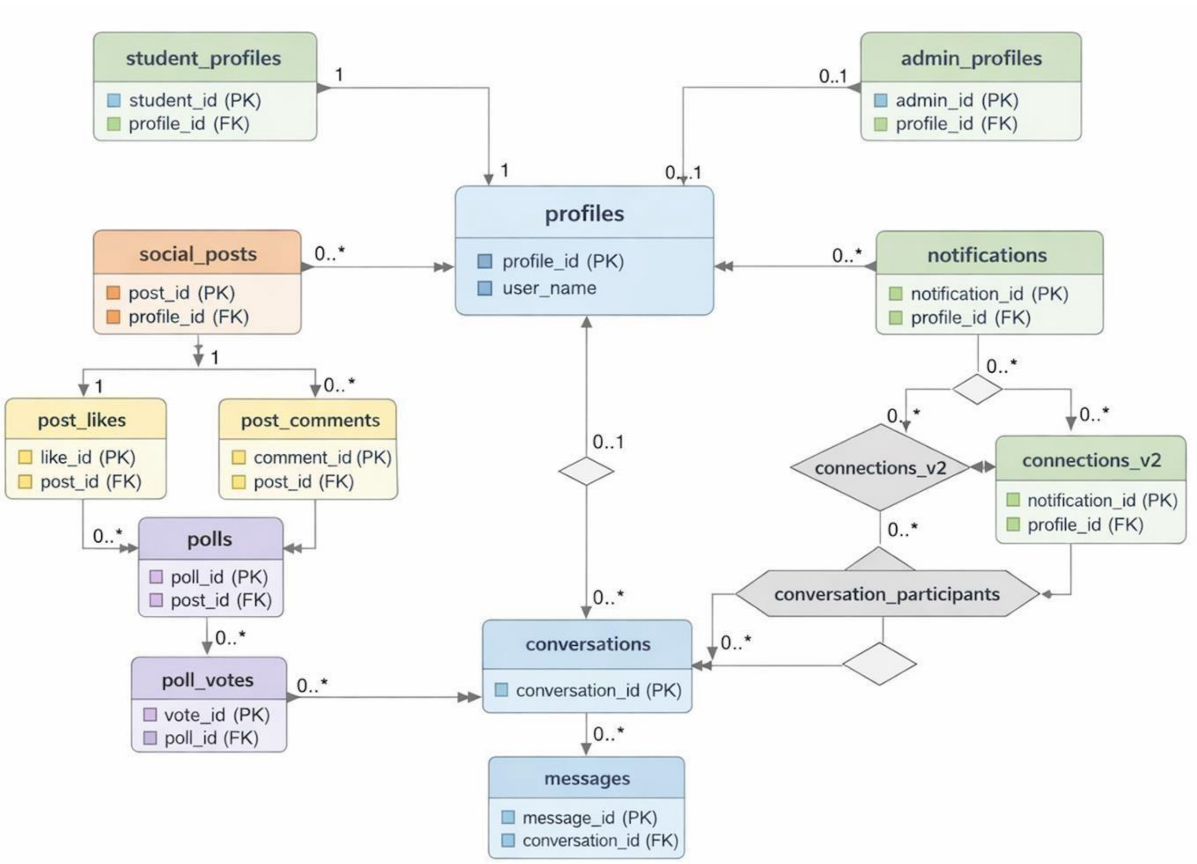

Figure 4.2 Entity Relationship Diagram

4.5. UML DIAGRAMS

4.6. Use Case Diagram

The use case diagram is a picture that shows how people talk to the system. It shows us how people use the parts of the system.The ChatPCCOE system has a use case diagram that shows what the Student, the Administrator and the ChatPCCOE system do when they talk to each other.The Student does things like register, login make posts send messages vote, in polls and manage friends.The Administrator does big picture jobs like send messages to everyone look at what the studentsre doing and check the records.

The ChatPCCOE platforms provides users with verification to verify what other individuals post. In order for the ChatPCCOE to work with maximum efficiency, the ChatPCCOE system must verify that any actions completed by students or staff are properly completed. The diagram outlines the actions a user must complete, such as signing into the system to complete an action, while other actions are optional (e.g., liking a post). It is easy to see from the ChatPCCOE use case diagram what actions a student and an administrator can perform with the ChatPCCOE computer application. The diagram is also helpful to understand the cooperation of the students and administrators when utilizing the ChatPCCOE.

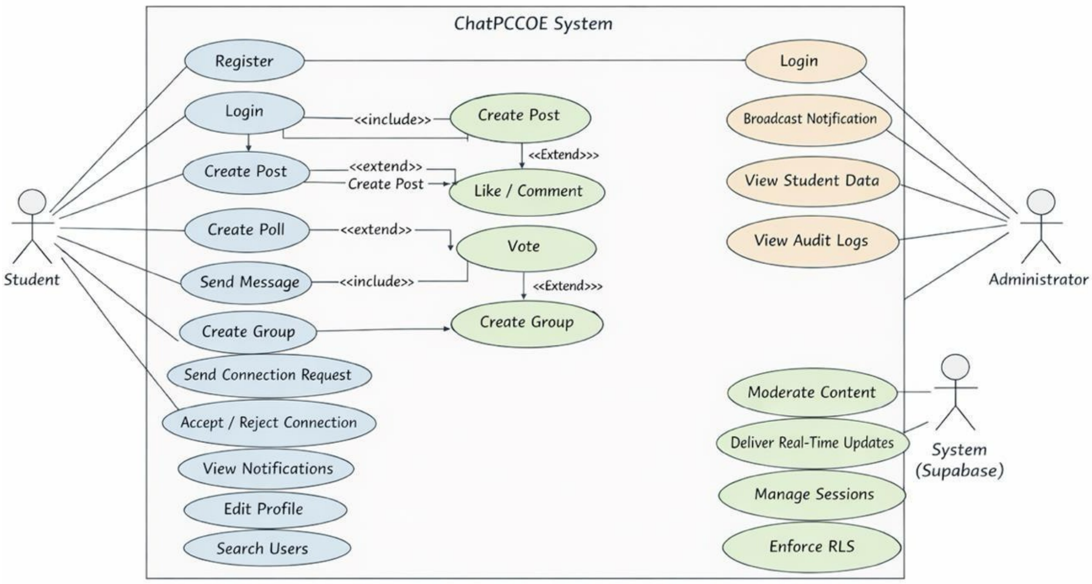

Figure 4.3 Use Case Diagram

4.7. Sequence Diagram

The ChatPCCOE messaging process will be depicted in a Sequence Diagram. The sequence diagram indicates how User A will communicate with User B using the Supabase backend and frontend applications.

First, User A will launch the messaging interface through the React application, which retrieves all of User A's conversations from the Supabase database. After that, User A can choose a specific conversation from those listed in the interface. Once User A has chosen the conversation, User A will be able to see all of the messages corresponding to that conversation. After sending his message to User B, User A will have his message stored within the Supabase database. When this occurs, the Supabase Realtime feature will perform some actions in response to providing User B with a copy of User A's message via WebSocket connection to the application; therefore, the interface of the application will also be refreshed to show the new message.

Eventually, User B will send a read receipt back to User A, allowing User A to know that User B has received the message in real time and have their message status updated. The ChatPCCOE messaging process is the communication between User A and User B.

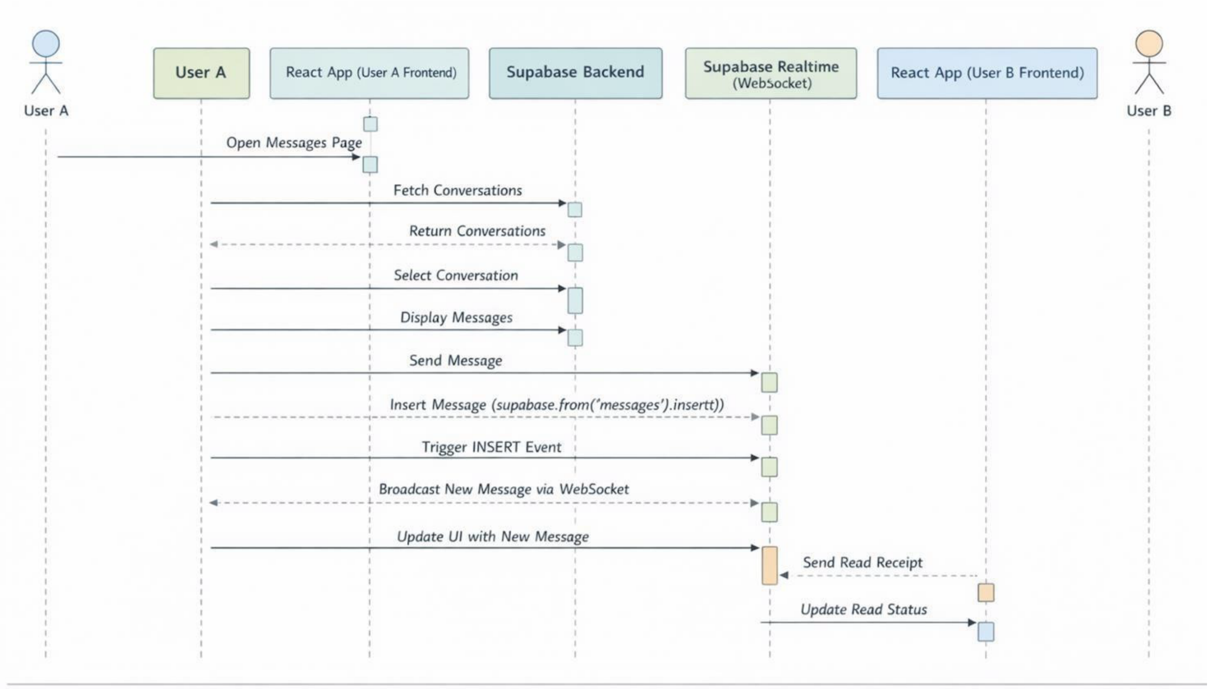

Figure 4.4 Sequence diagram

4.8. Component Diagram

In the ChatPCCOE component diagram, we see how the overall structure of the frontend application is laid out. The first part of the diagram is called App.tsx, which is responsible for managing the function of the entire application (i.e., it acts as the root component). App.tsx connects to AuthProvider, which provides functionality for ensuring that users have properly authenticated and keeps a record of all users' application-related information.

Routing layers act as a "map" for users to navigate between pages in the application and use something called Animated Routes to create a way to navigate between many of the pages within ChatPCCOE, such as: Home, Messages, Connections, Notifications Profile, Settings and AdminDashboard. Each of these pages consists of one or more individual components. Example: The Home page includes components such as SocialFeed and SocialPost; component may also contain one or more components.

The Messages page of the application is designed using a special component called MessagesLayout, which consists of a Sidebar, and a Container to provide the user experience of chatting with someone else. Additionally, there is another component called Service Layer, which manages the application state and logic using different technologies, including but not limited to: AuthContext, Zustand store and custom hooks.

The configuration of ChatPCCOE has been carefully designed to allow for the addition of more features or the transformation of how ChatPCCOE works relative to one another. In addition, the ability to visually represent an application's dependencies makes it much easier to maintain and scale. Furthermore, separating all the different components from one another and showing how they connect is critical to being able to produce a valid component diagram for ChatPCCOE.

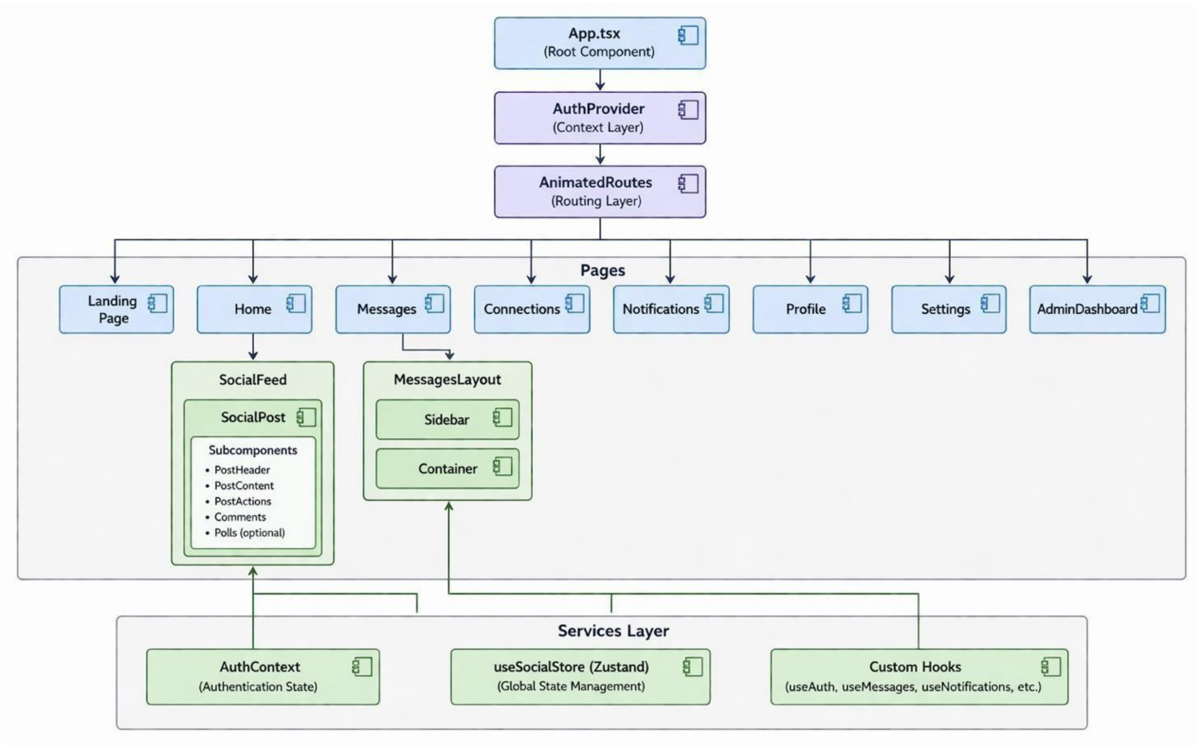

Figure 4.5 Component Diagram

5. PROJECT PLAN

5.1. PROJECT COST ESTIMATE

Cost estimating for projects is a crucial part of project planning. It helps determine how much money is needed to complete both the development and deployment of the proposed system. The development of the ChatPCCOE project is relatively inexpensive since it uses cloud-based applications and open-source technologies to meet its needs.

The ChatPCCOE project leverages Supabase to provide a Backend-as-a-Service, which is at no cost to us and provides all necessary elements such as database and storage capabilities. Thus, we do not need to build a backend infrastructure and will save a significant amount of money.

Furthermore, the development tools being used, Visual Studio Code and GitHub, are both free, which further contributes to keeping costs down. The hosting and deployment of the Lovable platform will include domain and hosting services, therefore these services will not require any additional expenditure.

Since the ChatPCCOE project is being developed at an academic institution for educational purposes, the only investment being made will be the developers' time and efforts—not monetary contributions. Many hours of work and effort are being put into this project by all members of the development team. No significant monetary expenditures will be incurred in association with this project.

Table 5.1 Project Cost Estimate

Item	Cost
Supabase Free Tier (Database, Auth, Storage, Realtime, Edge Functions)	$0/month
Lovable Platform (Development and Deployment)	As per subscription
Development Tools (VS Code, Git/GitHub, Browser DevTools)	Free
Domain and Hosting	Included in Lovable
Total Estimated Project Cost	Minimal (primarily developer time)

5.2. RISK MANAGEMENT

Risk management plays an important role in project planning. In order to develop a plan to avoid potential risks, we must investigate possible troubles that may arise, review the consequences of those troubles, and then formulate a strategy to prevent those troubles from occurring. ChatPCCOE's use of proper risk management will help to ensure that its systems remain safe and function as designed during periods of uncertainty.

ChatPCCOE uses user data to provide timely communication, utilizing internet/cloud- based services, so it is important to consider potential problems that might arise during the construction/installation of ChatPCCOE.

Our risk management process is a step-by-step approach. First, we identify the risks, then we assess how they would impact us and develop methods for minimizing their impact on us. By completing an in-depth analysis of each risk, it is possible to prioritize those with the greatest potential impact and allocate sufficient resources to effectively address the risks during the development of ChatPCCOE.

ChatPCCOE will continue running smoothly by managing risks through risk management plans, which should help prevent many things from going wrong. Since risk management is crucial to the ongoing operation of ChatPCCOE and minimizing issues, therefore we must manage the risk to ensure that ChatPCCOE operates securely and effectively..

5.3. Risk Identification

The process of identifying risk is about finding potential issues that may macth negatively on a system (ChatPCCOE). During the process of identifying risk, we found several risks related to the technology and manner in which ChatPCCOE works.

One of the more significant risks is a data breach, which occurs when individuals outside of ChatPCCOE access student information such as their PRN number, email address or telephone number. Another identified risk is API rate limiting. This occurs when the usage limits provided by Supabase are fully used up.

Another identified risk is the potential loss of an internet connection. The ChatPCCOE system uses a technology called WebSocket to provide a real-time streaming function to ChatPCCOE. A dropped internet connection would negatively impact the ability to retrieve real-time updates from ChatPCCOE.

One additional risk that we identified includes user moderation bypass. This is the act of trying to post content through the use of multiple word filters. In a similar vein, using a service such as Supabase presents a risk; if their systems fail, we are also impacted. Finally, if we do not validate that each user input is secure, we may open ourselves to XSS attacks which could compromise the entire system.

5.4. Risk Analysis

Risk Analysis means assessing risks, evaluating the severity of each risk and determining their likelihood of occurrence. This assessment enables companies to rank risks in order of importance, enabling the implementation of risk management strategies with the greatest impact. The purpose of risk assessment is to allow a company to perform risk analysis as part of its systematic approach to ensuring

compliance with applicable laws and regulations while reducing risks. The assessment of each risk allows for identification of ways to mitigate individual risks; therefore, it is critical to have an accurate listing of all potential risks and prioritize them based upon their respective impacts and likelihood of occurrence.

Table 5.2 Risk Analysis Matrix

Risk ID	**Impact**	**Probability**	**Priority**
R1: Data Breach	High	Low	High
R2: API Rate Limiting	Medium	Medium	Medium
R3: Connection Drops	Low	Medium	Low
R4: Moderation Bypass	Low	High	Medium
R5: Service Outage	High	Low	Medium
R6: XSS Attack	Medium	Medium	Medium

The examination makes it clear that a data breach poses a significant threat to our organisation and therefore, we must consider it an ongoing concern even though such an event is not likely to happen very frequently.A data breach occurs when an unauthorised person gains access to confidential information.This means we need to be aware of two more risks, both of which are also in the middle range - API rate limiting and XSS attacks.The analysis on risks to the organisation allows us to remain vigilant towards preventing breaches and mitigating their occurrence.

5.5. Overview of Risk Mitigation, Monitoring, and Management

Mitigating risk is all about minimizing the impact and/or the probability of experiencing a risk. At ChatPCCOE, a number of techniques have been implemented to ensure that the system is both reliable and secure.

For example, in preventing data breaches, ChatPCCOE uses Row-Level Security (RLS) on all database tables so that users only have access to view their own

information. Security functions are also used to limit the actions of users. Additionally, the administrators' actions are logged in order to record their actions for security purposes. In order to efficiently process large amounts of requests made to the system, they utilized improved query performance through both query optimization and caching to reduce duplicate requests.

When there are connection problems related to timing, ChatPCCOE utilizes the built- in re-connect feature from Supabase, in addition to conducting content checks at the server level. In the future, they plan on utilizing AI to perform content checks. In the event that the system does fail to function as intended, they have built-in error handling techniques and will be able to inform users of the status of the system. In order to prevent Cross-Site Scripting (XSS) attacks, ChatPCCOE cleans up any data entered by users prior to allowing it to be processed in order to eliminate the chance that any malicous scripts or codes can be submitted.

Monitoring the system is equally essential. For example, boards and logs provide insight into the overall health of the system and identify potential issues before they escalate. Early detection enables prompt resolution of identified issues.

5.6. PROJECT SCHEDULE

Project Scheduling is like a Time Table or Plan that shows us when and the order of things to be done in order to complete the overall Project. The Project Schedule helps ensure everything is completed and done in an orderly manner. The ChatPCCOE Project has a well defined and structured schedule. We break down the Work into Work packages by portions of the Work that need to be done for the overall Project (ex. Design, Produce, Test, and Distribute).

The Project Schedule aggrandizes how we utilize our Time and Resources in the most efficient manner possible. Each Work Package is assigned a time duration for completion, and we know the order and dependency of Work Packages. This allows us

to monitor Progress and provide for the timely completion of the Project. Project Scheduling is essential for the successful execution of the ChatPCCOE Project.

5.7. Project Task Set

Table 5.3: Project Task Set

Task No.	Task Description	Duration
1	Requirements Analysis and System Design	2 weeks
2	Database Schema Design and Supabase Setup	1 week
3	Authentication Module Development	1 week
4	Profile Management Module	1 week
5	Social Feed Module with Real-Time Updates	2 weeks
6	Content Moderation Edge Function	1 week
7	Real-Time Messaging Module	2 weeks
8	Group Chat Management	1 week
9	Connection Request System	1 week
10	Admin Dashboard	1 week
11	UI/UX Refinement and Responsive Design	1 week
12	Integration Testing and Bug Fixes	1 week
13	Deployment and Documentation	1 week

Each task is an important part of the system. We figure out how long each task will take based on how it is to develop. The tasks are done one, after the other. Repeated as needed following the Agile way of doing things.

5.8. Timeline Chart

The timeline chart, also known as a Gantt chart, provides a visual representation of the project schedule. It shows the start and end times of each task and helps in tracking progress.

Figure 5.1: Timeline Chart (Gantt Chart)

Task	W1-2	W3	W4	W5	W6-7	W8-9	W10	W11	W12	W13	W14	W15	W16
Requirements & Design	■												
Database & Supabase		■											
Authentication			■										
Profile Mgmt				■									
Social Feed					■								
Content Moderation					■								
Messaging						■							
Group Chat							■						
Connection								■					
Notifications									■				
Admin Dashboard										■			
UI/UX Refinement											■		
Testing												■	
Deployment													■

5.9. TEAM ORGANIZATION

Team structure is the way that the team can execute; it identifies roles and responsibilities of individuals on the team. Once the roles have been identified, it is possible for the team to work efficiently and complete tasks accurately. At ChatPCCOE, they create teams based on individual abilities and what is required to complete the project requirements. By doing these two things, efficiency will be maximized.

Each team member is assigned responsibilities that are aligned with their skills and what they are good at doing. The user interface looks aesthetically pleasing and functions correctly due to the efforts of the Front End Lead. By developing user experiences (UI), the Front End Lead ensures that users have a positive experience when using the developed application. The data layer and backend services will be developed and validated by the Back End Lead.

A Feature Developer will add the various functional elements (i.e., the social feed, messaging, notifications, etc.) of the application. The Feature Developer will ensure that these features function properly, therefore enhancing overall project success.

ChatPCCOE has designed their development team to be successful by creating a collaborative environment amongst the Front End Lead, Back End Lead, and Feature Developer, which produces an overall successful product.

Table 5.4: Team Organisation

Team Member	PRN	Role / Responsibilities
Mangal Singhal	122B1D058	Feature Developer – Social Feed, Messaging, Connections, Notifications
Anshul Wagh	122B1D066	Backend Lead – Database Design, RLS Policies, Edge Functions, API Integration
Vedant Sonawane	122B1D060	Frontend Lead – Authentication, Landing Page, Profile Management

The team works together. Talks to each other often. They make sure they are doing their tasks in a way that fits with what othersre doing.They use methods, like giving updates and having feedback sessions.This helps make sure the project development goes well.The way the team is organized helps them finish the project successfully.

6. PROJECT IMPLEMENTATION

6.1. OVERVIEW OF PROJECT MODULES

The CHATCCOE is comprised of small components that play vital parts and are all interconnected to create an environment that allows users to interact with other users; to share communications; and to learn from each other.

All of these components contain different features such as creating accounts, checking in on contacts, sending messages, establishing groups for chatting, managing contact lists, being notified when someone has communicated with them, and maintaining the integrity of the system's components. They all communicate and interchange data in a manner that is logical and provide the end-user with a pleasant experience; all of which have been designed and built in such a way that they use Supabase as a backend storage mechanism, enabling all components of the system to work together efficiently, and ensuring that all system data is current and accessible.

Module 1: Authentication and Authorization

The user identity and access control part of the system is the authentication and authorization module which does a variety of functions like user registration, login, and session management. The module also utilizes role-based authorization which determines what users have access to and what they cannot have access to.

Supabase authentication services are used for email-based logins, which utilize JSON Web Tokens (JWTs) for managing user sessions.

An important feature in the authentication and authorization module is AuthContext, created using the React Context API. This is a wrapper around the application and provides a way for users to access authentication-related information from anywhere in the application. This includes the user, their session, their role in the system, and how they authenticated.During user registration, the system validates the data entered by the user. Using roles to define the permissions for users within the application will ensure that only users who are authorized will be able to access those features within the application. For example, a student will be given different access to the application than a faculty member will be given and will have different access than what a system administrator will have. The application also includes security controls, such as session and token security, that help to protect data from anyone who should not have access to it (to prevent unauthorized users from accessing the application). The importance of the Authentication and Authorization Module(s) lies in the fact that they both maintain the overall security of the system and the ability to use the Authentication and Authorization Module(s) provides for the continued security of the system.

Module 2: Social Feed

With the Social Feed module, users can express themselves as well as see and communicate with other people on the module. To do this, we use a technology called Zustand, which keeps track of everything happening in the Social Feed module in a simple and fast manner.

In addition to displaying posts, allowing users to like/dislike posts, comment on posts, and vote on posts, the Social Feed module performs numerous other tasks. The social feed database contains a list of posts created by users, social_posts. When displaying posts, we will also retrieve information about the user who created the post.

With CreatePost, users can create three types of content: write a post, upload an image or video, or create a poll.

Keeping the Social Feed up-to-date is critical; to do this, we use Supabase real-time subscriptions to listen for changes made to the Social Feed database. When a user makes a post, likes a post, or comments, everyone viewing Social Feed will see the change at the same time. This will provide a fun real-time experience for the user.

Module 3: Real-Time Messaging

With this messaging system you can communicate without paying for an app. Messages can be one-on-one or in a group context. The system allows users to send and receive messages and view when their message has been read.

There are two helpers: useConversations and useMessages. The useConversations helper takes care of the user's contact list and allows the creation of new conversations. The useMessages helper retrieves, sends and updates message information in real-time. The user can send all types of message: text, images, video, audio/voice files, and PDFs.

Files uploaded by users are stored in Supabase Storage. Supabase Storage will make sure that the user's uploaded files are safe and that they do not consume too much storage space. Messages in the messaging system are received via Supabase Realtime.

Therefore, messages will be received instantaneously and the system will always be updated. Using this messaging system, it is extremely easy for users to communicate.

Module 4: Group Chat Management

A group chat management system allows users to create and manage groups. To create a new group, you use the CreateGroupDialog and to manage the operations of a group, you use the groupService.

Within this system, there are many capabilities available to manage each group such as: allowing/disallowing members from being part of a group, creating each member's role, and managing what a member can do in a group(s). Row-Level Security policies are used (via Supabase) to restrict access to various functions based upon a person being the group's administrator or lead.

The group chat system allows for collaboration via the ability for everyone in a group to communicate within one interface. This is important when it comes to school related topics, working on group projects, and planning events. The group chat system can be used as a communication method to complete tasks. In essence, group chat can be very beneficial for group discussions.

Module 5: Connection Management

The connection management system provides users with the opportunity to connect with someone in a manner similar to how professional websites provide users with capabilities to connect with others (by allowing you to send or receive connection requests, as well as cancel them).

The part of the system used to manage connections is a database table called connections_v2 which tracks all of the connections between individuals, using a connection_status field to determine the current state of the connection. When a user accepts or rejects an invitation, the connection management system uses a process known as Remote Procedure Calls (RPC) to ensure that any actions related to the connection are processed correctly, and that data remains consistent throughout.

The connection management module provides students, faculty and alumni the ability to connect with one another through the connection management portion of the system. In addition to providing the ability to connect with another individual, the connection management module ensures that the integrity of the connection records are maintained by not allowing invalid connection states to exist. These processes are performed in order to maintain a high level of accuracy and security for the connection management module and for the integrity of the data being stored within the connection management module.

Module 6: Notification System

By using the notification system, users are provided with information regarding happenings within the platform. Each user receives common messages, as well as the specific communications they receive (i.e., Friend Requests), through the Notification System.

The useNotifications Hook manages all types of notifications and organizes notifications by when a notification occurred. All notifications are updated in real-time via the use of Supabase subscriptions, so that users can receive notifications immediately upon being sent.

Additionally, the notification system is important because it provides everything necessary to keep users engaged, and will assist users in staying up-to-date with regards to both the notification system and the overall platform itself.

Module 7: Content Moderation

To prevent inappropriate or offensive content from being uploaded to the platform, we utilize a content moderation module. This module checks for prohibited/negative content in two ways.

The first way in which we check for profanity and offensive language and words is on the users computer (before anything is sent to the server) and the second way is after

the input has been sent to the Supabase Edge Function. Upon being sent to the Supabase Edge function, the input will be checked against a list of prohibited words.

If the Supabase Edge Function finds any prohibited words in the user's input, it will prevent the post from being made and display a message to the user indicating that there was an issue. The content moderation module is very important because it ensures that our users can use the platform in a professional manner.

6.2. TOOLS AND TECHNOLOGY USED

ChatPCCOE's system is built upon a stack of technologies that maximize efficacy and scalability. Users' interactions through a web interface have been constructed using React.js, with TypeScript as the scripting language. The underlying infrastructure (i.e., the database) is served by Supabase, which manages access control (identifying and allowing valid users) and provides real-time functionality to users. Therefore, the ChatPCCOE system leverages Supabase as its backend and React.js as its frontend.

Table 6.2 : Tools and Technology

Category	Technology	Purpose
Frontend Framework	React 18.3	Component-based UI with hooks
Language	TypeScript 5	Type-safe development
Build Tool	Vite 5	Fast builds with HMR
CSS Framework	Tailwind CSS v3	Utility-first responsive styling
UI Components	shadcn/ui	Pre-built accessible components
State Management	Zustand	Lightweight global state
Animation	Framer Motion	Page transitions and animations
Routing	React Router DOM	Client-side routing (11 routes)
Date Formatting	date-fns	Date formatting utility
Validation	Zod + React Hook Form	Schema validation and form management
Toast Notifications	Sonner	Success/error toast messages
Icons	Lucide React	Icon library
Database	Supabase PostgreSQL	Managed DB with RLS
Authentication	Supabase Auth	Email-based auth with JWT
File Storage	Supabase Storage	Object storage for uploads
Real-Time	Supabase Realtime	WebSocket subscriptions
Serverless	Supabase Edge Functions (Deno)	Content moderation, avatar upload
Package Manager	Bun	Fast JS package manager

6.3. ALGORITHM DETAILS

6.4. Content Moderation Algorithm

The content moderation algorithm looks for and prevents users from posting content. It does this by examining the text written by users to see if they contain any prohibited words.

The content moderation algorithm begins by creating a list of all prohibited words. Each word on this list has a different representation that is validated to allow it to function properly with the content moderation algorithm and is then combined into a super-rule. The content moderation algorithm uses super-rules to evaluate the submitted text and if it does not produce any recommended actions to prevent the submission of the original text, it will permit the submission of said text.

The content moderation algorithm will therefore allow updates to be identified quickly and will not reduce performance on the platform due to having to consider any prohibited terms that are present within the submitted text. The content moderation

algorithm operates quickly since it does not take long to assess the content submitted, which means it is an ideal tool for real-time situations such as the content moderation algorithm.

6.5. Connection Request State Machine

Throughout the lifecycle of a connection request between users, the Connection Request State Machine manages the entire process to ensure consistency and to prevent errors.

There are four possible connection states; No Connection, Pending, Accepted and Rejected. Each transition, between these states can occur through sending a request to create a connection (request), accepting a request or cancelling a request. Each transition is validated based on the state of the connection request before completing the action.

This mechanism provides each user with a valid way of monitoring their relationship with other users and will prevent the same connection request being sent more than once. Each request is properly logged by the Connection Request State Machine.

6.6 Real-Time Subscription Management

When data changes in the database, users will receive notification of the change by using the subscription algorithm. The update process uses Supabase Realtime, which listens for database events.

When a user visits a web page that needs real-time updates, the system creates a WebSocket connection to the relevant channel(s) and subscribes the user to receive real- time updates. The user receives a notification when the data in the database changes. The user can update their current view on the web page based upon the notification received.

7. SOFTWARE TESTING

7.1. TYPES OF TESTING

Testing ChatPCCOE software is a very important part of making sure that ChatPCCOE works correctly, is secure and does what ChatPCCOE was designed to do. By following an organized process and to ensure that we used each component, looked for issues, and verified that it met all the requirements established for ChatPCCOE.

To complete the above mentioned tasks, we completed multiple types of testing for ChatPCCOE. This was completed by testing a component individually before testing how all components work together, testing the capabilities of ChatPCCOE, testing the security of ChatPCCOE, testing how ChatPCCOE displayed on different devices and testing how effectively ChatPCCOE would function in a live environment.

Each form of testing (i.e., part of ChatPCCOE or the entire ChatPCCOE system) included specific aspects of verification that allow us to confirm the accuracy of an individual part of ChatPCCOE and the overall system of ChatPCCOE. The way we tested ChatPCCOE was meant to provide us with assurance regarding the ability of ChatPCCOE to be correct and reliable so that ChatPCCOE will perform as expected for the end user in any circumstances that ChatPCCOE might be used for.

Unit Testing

Unit Testing refers to putting parts of something together in order to ensure that each piece will function properly. The team at ChatPCCOE tested utility functions (methods) and services individually to check that they are operating correctly. Some examples of methods that we tested were: "sanitizeInput", "validatePassword", "validatePCCOEEmail", and "checkRateLimit". While performing this testing, we put various types of input through these functions to determine whether or not they were validating the input as intended and then handling any errors that occurred.

Independently testing the Edge Function for content moderation also required a unit test

to ensure that this Edge Function can identify and filter out undesirable content. The unit testing process is beneficial because it identifies potential issues within the system early on, making it easier to identify and correct problems later. When the entire system consists of working units, the reliability of the overall system improves significantly. Testing several units of code: including, but not limited to the Edge Function for content moderation as well as the other utility functions, will provide us with an assurance that ChatPCCOE will function in an expected manner.

Integration Testing

Integration testing is a way to determine how components communicate with each other and to test the overall functionality of the integrated system.

At ChatPCCOE we tested to ensure proper integration of the Frontend and Backend components of the Application using a full end-to-end process and did this through functional testing of the application as the user signed up via the enrolment process, created entries in the database (entire flow through the application) and message users and manage their user connections (connections between users).

An example of testing through the Application workflow was an enrolment process, (the user) is able to successfully sign up (enroll) on the ChatPCCOE system and that the Application has validated everything that the user has entered when they signed up, that the Application's security process has assigned a Login and Password for the user and that the user's profile information has been stored in the database and is retrievable.

Also, testing was also conducted to ensure that users can send and receive messages from one an other in real time.

Integration testing is essential in order for us to determine the overall functionality and performance of ChatPCCOE as an integrated system and to validate through testing the integration of ChatPCCOE with its integrations.

Functional Testing

Functional testing checks if the system does everything it should. We accomplish this by looking at system requirements. In the case of ChatPCCOE, we checked for the functionality of many things such as authentication, profile management, feeds, messaging,

We test every piece of the system to ensure it works correctly in a normal state. For example, we would check that users could post, communicate with one another and send messages and receive notifications. We wanted to ensure that the system was functioning correctly and that the users could achieve what they needed to do with the system. There is reason for concern around functional testing because it is an assurance that the system was functioning as expected and that users will be satisfied with ChatPCCOE’s features. We checked all of ChatPCCOE’s features for the functionality of all of them.

Security Testing

ChatPCCOE's security is very valuable because there's a lot of personal user data associated with this app. Therefore, we've verified that everything in place to secure user information against unauthorized users is working as designed. We also verified the proper operation of Row-Level Security (RLS) policies by testing user access to multiple people's data without permission. This is to confirm if RLS policies work the way they're intended.

In addition, we also made sure that Cross-Site Scripting (CSS) attacks could not be executed against the app. We conducted a number of very high-volume testing scenarios to see if the app was able to prevent abuse by blocking incoming code and/or web requests.

All of these tests provided a solid confirmation that ChatPCCOE is secure, and that the app can mitigate any web issues that may arise. This security testing of the ChatPCCOE application is part of ensuring that everything in the application works as designed.

Responsive Testing

There are lots of different combinations of devices and screen sizes to test for how well the application works across many types of device. We tested ChatPCCOE on numerous devices, including many desktop computers and tablets; through both of which we tested using mobile phone size (smartphone) resolutions. The mobile resolutions tested included 1920x1080, 1366x768, 768x1024, and 390x844.

We checked how well the user interface of ChatPCCOE works on all devices by checking the user interface of ChatPCCOE and determining if the user interface of ChatPCCOE appears correctly, looks good and is easy to use. We used Tailwind CSS, a web-based design framework, with ChatPCCOE to ensure ChatPCCOE is completely compatible with all devices.

This testing has an important goal; that goal is to ensure that every potential user of ChatPCCOE can have an easy-to-use interface to access software via ChatPCCOE and therefore provides value, for those users, by successfully completing the tasks necessary to complete to successfully interact with the application via the account associated with their user of ChatPCCOE.

Real-Time Testing

To determine if Supabase Realtime was actually functioning, tests were done with Supabase Realtime's live updates. To test functionality, many browser tabs were opened and actions taken including sending messages, posting comments, and receiving notifications. Tests confirmed that all tabs received their updates in an interval of less than 1 second, with all being updated within 500 ms, allowing for a real-time, dynamic, and interactive user experience when using Supabase Realtime. Testing of this sort is crucial for live updates involving messaging and/or notification systems on Supabase Realtime since those utilizing the system require up-to-date data.

7.2. TEST CASES AND TEST RESULTS

To verify proper operation of the system, we create test cases that allow us to view results as well as compare them against what we expect. Each test case will provide a unique identifier, a description of what has been tested, and the input given when testing it, the expected result when testing it, and finally the outcome of whether or not testing worked as expected on each individual test case. For example, with ChatPCCOE, there are several test cases that correspond to the features that exist related to logging in, creating content, messaging, and maintaining security as well as performance testing.

Table 7.1: Test Cases and Test Results

TC ID	Description	Input	Expected Result	Status
TC-01	Student Registration with Valid PCCOE Email	Valid PCCOE email, strong password, PRN	Registration succeeds, profile created	PASS
TC-02	Registration with Non-PCCOE Email	email=john@gmail.com	Validation error, registration blocked	PASS
TC-03	Registration with Weak Password	password=123	Strength indicator shows weak	PASS
TC-04	Create Text Post with Clean Content	content=Hello everyone!	Post created, appears in feed	PASS
TC-05	Create Post with Profanity	Content with profanity	Post rejected, error toast shown	PASS
TC-06	Send Connection Request	User A sends to User B	Record created, B gets notification	PASS
TC-07	Accept Connection Request	User B accepts from A	Status updated, A gets toast	PASS
TC-08	Send Real-Time Message	User A sends Hello to B	Message appears in both within 500ms	PASS
TC-09	File Upload in Chat	User attaches image	File uploaded, message with file_url	PASS
TC-10	Admin Broadcast Notification	category=Exams, title=Midterm	Notification appears for all users	PASS
TC-11	XSS Attack Prevention	content=<script>alert(1)</script>	HTML entities escaped, script blocked	PASS
TC-12	Rate Limiting	6 rapid login attempts	6th attempt blocked	PASS

TC-13	Unauthorized RLS Access	User A reads B's messages	Empty result set returned	PASS
TC-14	Create Group Chat	name=Project Team, 3 members	Group created, creator as lead	PASS
TC-15	Poll Creation and Voting	Poll with 3 options, 1 vote	Poll created, vote counted	PASS

Analysis of Test Results

All parts of ChatPCCOE are tested to work correctly according to what they are intended to do. Testing also confirms chatPCCOE's safety and speed when handling things on time. No tests were conducted that would cause chatPCCOE to fail, so the builders of chatPCCOE did a good job of building it and performing tests successfully as well. Multiple situations were tested by chatPCCOE to show it is very reliable on not breaking by conducting a mixture of testing for chatPCCOE. ChatPCCOE is, therefore, good to go for anyone who would like to use ChatPCCOE

8. RESULTS & DISCUSSIONS

8.1. OUTCOMES

ChatPCCOE provides a platform for individuals to communicate with others on campus in real-time. The primary objective of this application was to provide a mode of communication for students, faculty, and staff so they could communicate with each other without the necessity of multiple tools such as text messaging applications, email applications, and informal groups.

Within ChatPCCOE, there is a feed component within the platform that allows users to create posts containing text, images, and video content. Polls can also be created. Once a user submits a post, all other users have the ability to view that post nearly instantaneously due to the use of WebSocket-based Supabase Realtime within the system. While sending posts, users will receive posts within 1/2 second of receiving. This makes using the platform enjoyable and keeps users engaged.

ChatPCCOE also provides a method for users to communicate via messaging with other users privately or group-wise. There are multiple ways users can send messages between other users such as text messaging, Pictures and Video. Users also have the ability to see if any other user has viewed their message. This application has simplified the way users communicate with each other and also provides a productive means for users communicating during group-based projects.

In addition, the platform includes parts that enable people to connect with others through a network. This feature allows people to request to be friends with someone else (who can either accept or decline the request). Users on the platform are notified of other user's requests to become their friend and of updates to their profile. The platform was built as an organized structure that is appropriate for a school setting.

The creators of the ChatPCCOE platform implemented safeguards to protect user privacy. All users have access to their own information only (as opposed to others'). There are also filters implemented on the platform that screen what words can be used in the system, thereby keeping all user exchanges friendly in nature. Additionally, the platform functions properly across all types of devices - including computers, mobile

telephones, and tablets.

The ChatPCCOE platform has successfully demonstrated how a school can establish an effective communication system using web-based technologies. This means schools can have a safe and effective communication system that is easy to use. The ChatPCCOE platform has proven that a systematic approach to developing an open communication system can be accomplished.

8.2. RESULT ANALYSIS AND VALIDATION

The efficiency and performance of ChatPCCOE were evaluated through a series of checks. Checks included how well the system works, how quick the system works, how safe the system works, and how easy the system works. The checks suggest that ChatPCCOE performs as intended and performs well under conditions.

One item evaluated was how well the system prevents bad content from being displayed. The system's rules consist of an inventory of 37 words with a defined set of rules that enforce full word matching against the inventory. This improves the accuracy of finding bad content and reduces false positives. Therefore, "assessment" is not flagged as a bad word when using this algorithm because the system performs full word matching against the inventory. This demonstrates that the algorithm is functioning properly. However, it is possible for users to disguise words by providing alternate character representations of those words and, subsequently, the system is not capable of identifying those words. The future of this system will address this issue by incorporating intelligence as part of the content check process.

The performance of the system was assessed via Supabase Realtime services. The Supabase Realtime service allows for updates to be made available to everyone at once (i.e. in real-time) and measured. The tests showed that updates are available in less than 300 ms. This is considered very low latency and fast.

The ongoing operations of the system when the internet connection is unstable also provide evidence that the system's overall operation and functionality is continuing as intended and that users are having a good experience.

Access and common threats to the security of the system were checked and verified. When attempting to access user data, no access was granted due to security measures in place. The ability of the system to support users and function reliably was also evaluated. Supabase services provide support for multiple simultaneous users on the system. The system's architecture allows for the addition of new features without affecting the current operation of the system. The user interface of the system produces consistent experiences on different devices and screen resolutions.

ChatPCCOE is user-friendly with a contemporary design, functioning on multiple devices and providing organized notification systems and layouts that allow easy finding of resources. Overall, ChatPCCOE generates high efficiency and security for a successful platform supporting its intended tasks. ChatPCCOE accomplishes its intended purpose for facilitating real-time interactions with proper content management and secured environments makes it ideal for application within academic environments. ChatPCCOE demonstrates dependability; therefore, ChatPCCOE is also an efficient and secure platform, essential for ChatPCCOE.

8.3. SCREENSHOTS

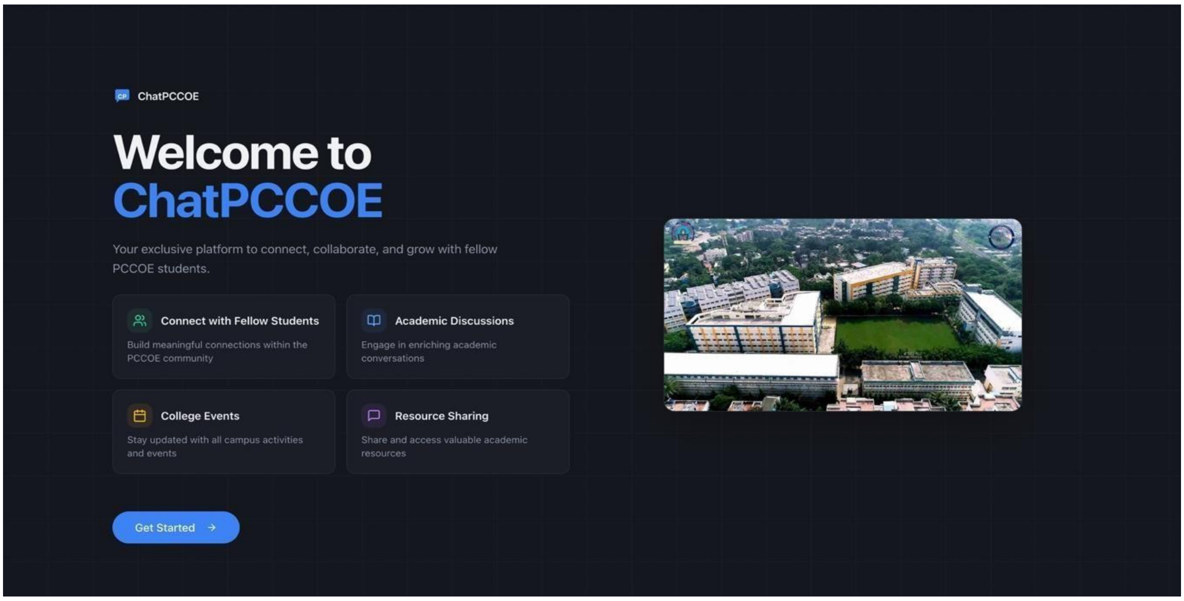

Figure 8.3.1 Interface of ChatPCCOE Platform

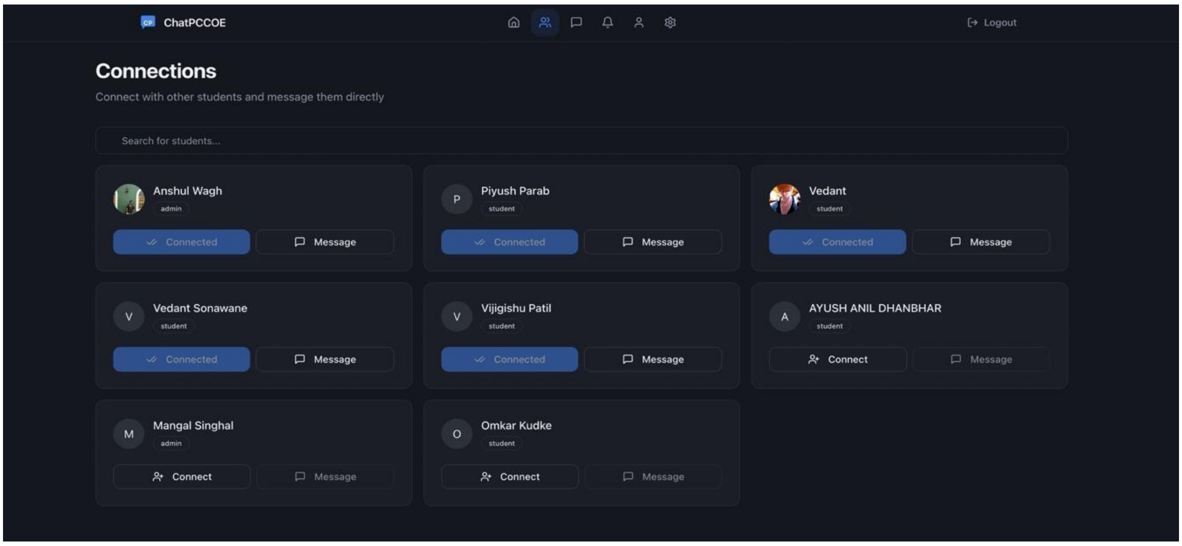

Figure 8.3.2 Student Connection Module of ChatPCCOE

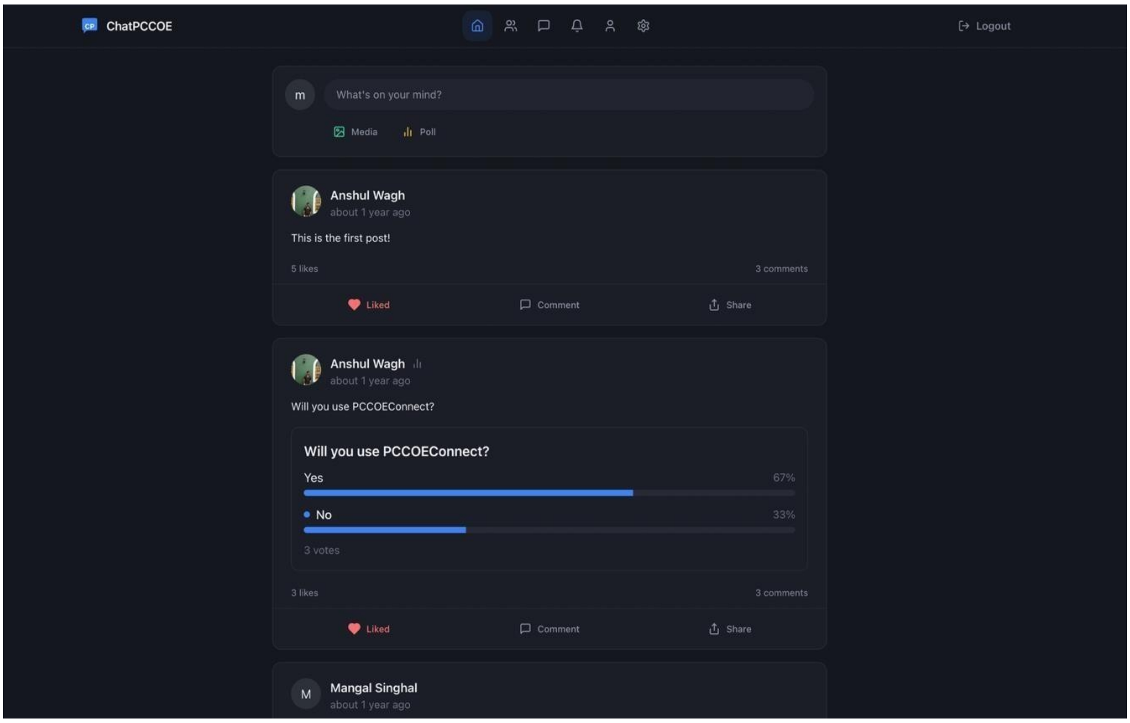

Figure 8.3.3 Social Feed and Interaction Module of ChatPCCOE

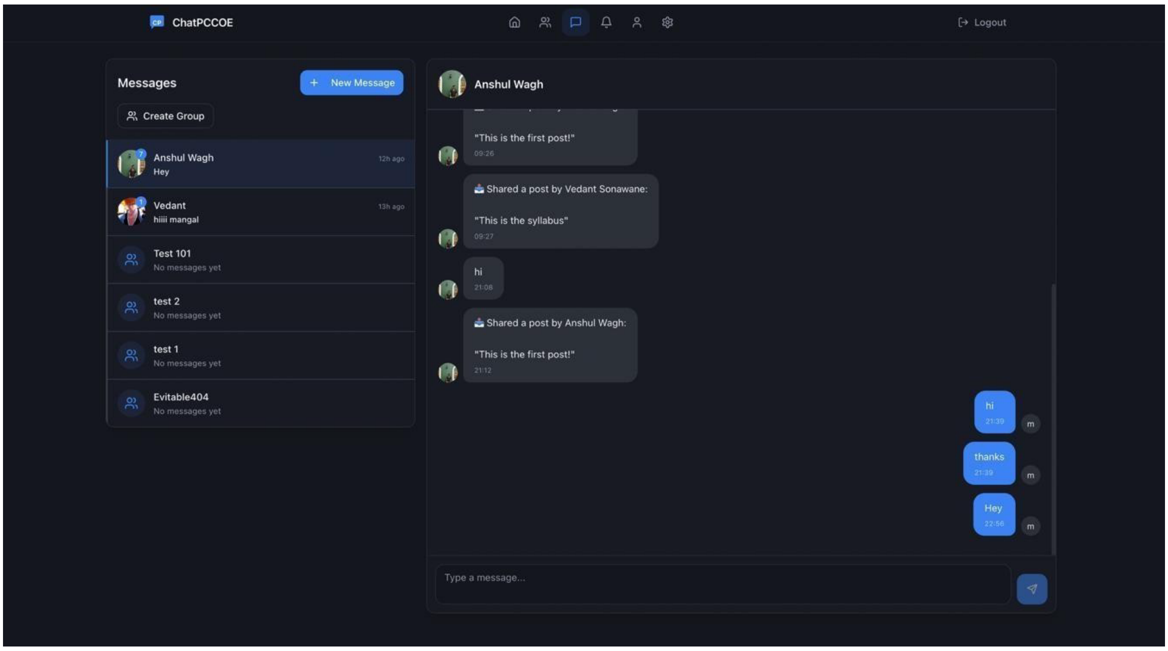

Figure 8.3.4 Real-Time Messaging Module of ChatPCCOE

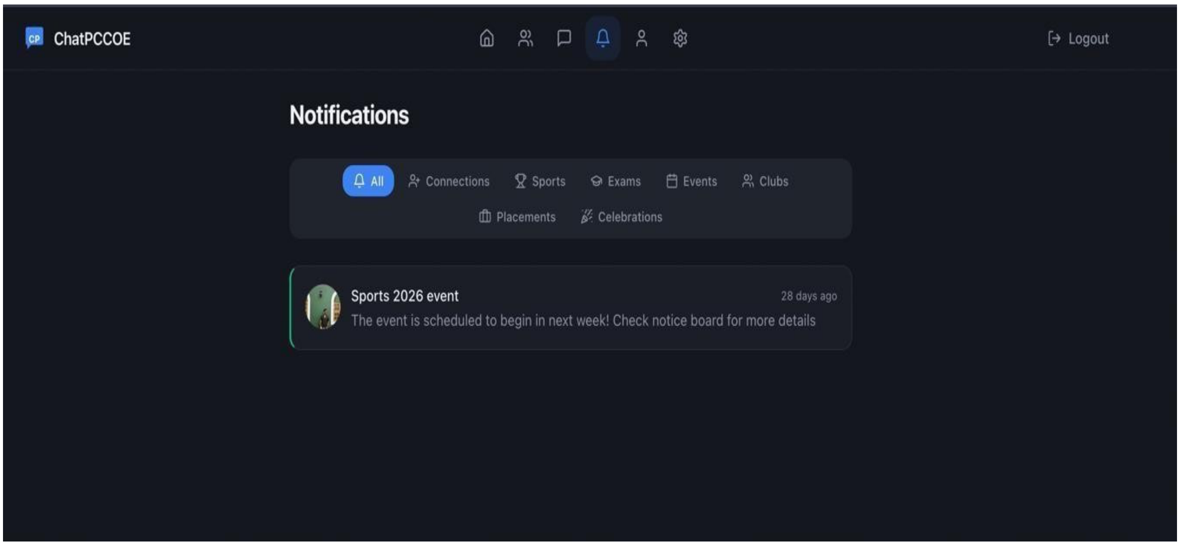

Figure 8.3.5 Notification Management Module of ChatPCCOE

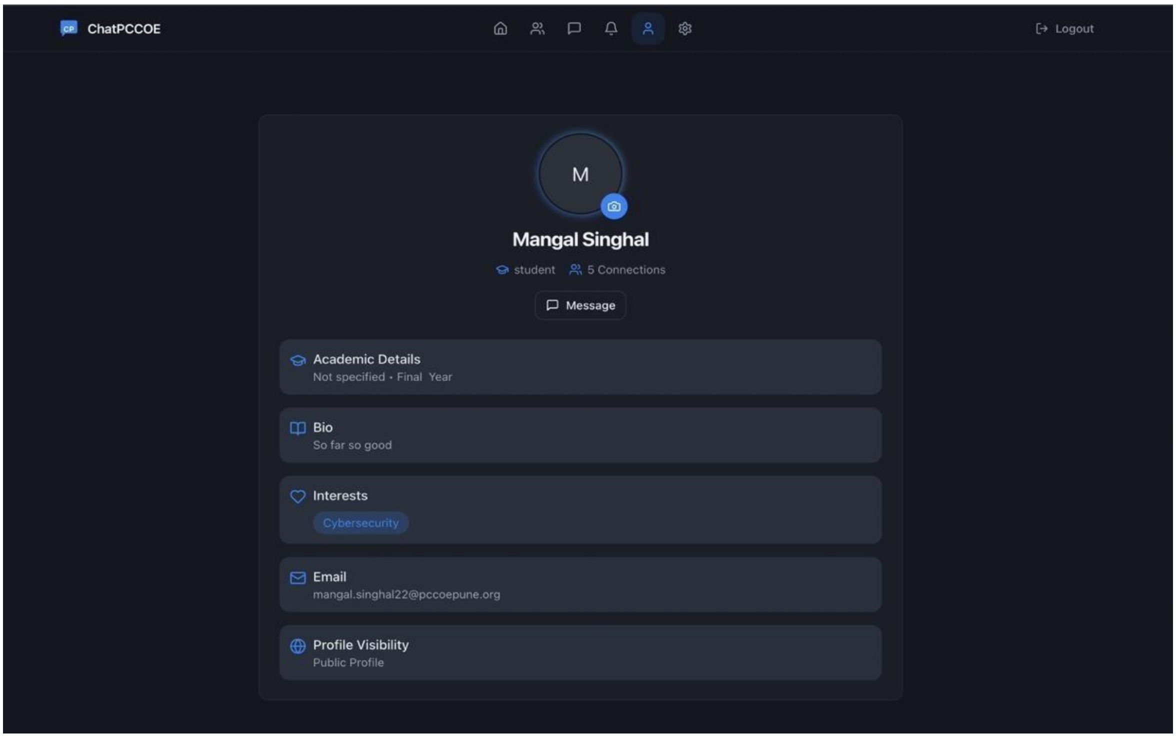

Figure 8.3.6 User Profile Management Module of ChatPCCOE

9. CONTRIBUTION TO SUSTAINABLE DEVELOPMENT GOALS

9.1. INTRODUCTION TO SDGs

The UN's Sustainable Development Goals were a collective commitment made by the UN in 2015 to solve the most critical issues facing our world, including issues such as extreme poverty, inequality, or lack of equitable rights for all humans on Earth, climate change, thoughts and actions regarding how we interact with our planet, access to quality education for all people, and the enjoyment of happiness and health. The Sustainable Development Goals serve as a guiding framework for governments, organisations, and individuals to create a more sustainable future for every citizen.

Computers and IT systems play a big role in helping achieve the UN Sustainable Development Goals both for schools and for educators. Technology has also enhanced education by facilitating cooperation, collaboration, brainstorming new ideas, and building stronger communities within education. Schools can also use technology (computers or the Internet) to improve educational content, make educational experiences more interactive and enable students to work collaboratively to create, share, and build stronger communities within the framework of education.

ChatPCCOE has partnered with the United Nations to promote Sustainable Development Goals and create an innovative, collaborative and educational community among students at the college level. ChatPCCOE creates opportunities to build relationships and connect; engages your learning experience to increase the enthusiasm of your peers to learn together; and ultimately to have a higher quality of life at school.

9.2. MAPPING OF THE PROJECT TO RELEVANT SDGs

ChatPCCOE helps many people access education by working with teachers and students. They build educational tools using TypeScript, React, and Supabase to provide a platform for teachers and students.

The platform is solving problems for both teachers and students There are many types of solutions they are providing to help individuals reach their full potential; supporting the achievement of the Sustainable Development Goals. PCCOE, Department of Computer Engineering (Regional Language) 2025-26 By working to enhance education around the globe, ChatPCCOE is helping to create a better world for all through its many efforts.

Table 9.1: SDG Mapping Detailed Explanation of SDG Mapping

SDG	Relevance to ChatPCCOE
SDG 4 – Quality Education	Ensures timely academic information delivery through categorized notifications (Exams, Placements, Events). Enables knowledge sharing, academic discussions, and collaborative learning.
SDG 9 – Industry, Innovation, and Infrastructure	Demonstrates innovation using React 18, TypeScript, Supabase, and WebSocket real-time communication to solve a real-world campus problem. BaaS with RLS is an innovative secure architecture approach.
SDG 11 – Sustainable Cities and Communities	Strengthens campus community through peer networking, group formation, and information sharing. Promotes community engagement via notification categories for Clubs, Events, Sports, and Celebrations.

ChatPCCOE enhances student educational attainment and makes it easier for students to access and find information through well-organized communication, which includes distributing organized announcements and providing resources to be shared among students. Through ChatPCCOE, students receive up-to-date information regarding examinations, academic events and/or activities or any other matter which may affect their education.

Furthermore, ChatPCCOE will allow students to collaborate in learning and will offer multiple ways for students to collaborate, including using the news feed as well as

through a variety of discussion type features. In turn, this provides students with opportunities to assist one another in completing school projects and results in a more pleasant and accessible educational experience.

ChatPCCOE is also providing assistance to Sustainable Cities and Communities. This is achieved by facilitating a school community where the students of the school can interact with one another, the staff at the school, and former students. ChatPCCOE provides the school with group chat capabilities, notification and event updates. These features will help build a cohesive community at the school with access to real-time information, which is necessary for a successful learning environment.

9.3. PROJECT IMPACT ASSESMENT

ChatPCCOE generates two different fields of effect; both affects how individual teachers/students respond, as well as collectively provide a level of impact to all of the students and faculty members of the entire college. Clearly, ChatPCCOE has a positive effect on the entire student population, teaching faculty, and overall college community.

Direct Impact

There is one company that assists students in all areas, enabling them to review any new office, group email, and other updates from one centralized location.This system can provide notification of new information on your device; this is true for exam schedules, job opportunities, or department functions, enabling you to look quickly for new items.

The use of this system provides a means for students to connect and interact with one another, developing collaborative teamwork and providing an avenue for collaborative projects and the ability to develop relationships with one another through using the information generated.

The rules set forth to establish a professional environment for respectful discussion will discourage inappropriate posts and help establish a place for professional, respectful discussion.

Indirect Impact

There are many advantages of using ChatPCCOE but there are some not so obvious benefits that users might not realize; for example, one such example of a non- obvious benefit associated with ChatPCCOE is its ability to reduce the number of tools required for individuals to communicate with one another. By having fewer tools available to communicate with, ChatPCCOE provides users with more confidence about whether or not the contents of their conversations will remain private; thus, reducing the likelihood of people accessing private information when it is stored in multiple locations. In addition, by providing other institutions with a model for using web-based technologies to create and implement systems that operate effectively and efficiently while offering maximum protection to the privacy of their users, ChatPCCOE will help them decide how to develop their systems in the same manner as ChatPCCOE has done.

The developers' use of Row-Level Security (RLS) to secure their database and limit access to records based on user permissions has created a stronger level of protection for the application. Developers should consider building applications with security in mind to improve their overall level of security and give others ideas for how they can do the same. ChatPCCOE does more than just help those who use it; it also improves education technology and improves the overall quality of design of systems used to deliver education.

9.4. CHALLENGES IN FUTURE SCOPE

Challenges Encountered

In the development of ChatPCCOE, we faced multiple technical obstacles that provided an idea on how to enhance ChatPCCOE. For instance, creating the role-based security (RLS) policy for ChatPCCOE presented us with a large hurdle. The recursive nature of the role-based security policy had loops in it so ChatPCCOE would perform slowly. We solved this issue by leveraging the use of SECURITY DEFINER functions to manage access to database operations via ChatPCCOE.

Another issue related to real-time subscriptions management in ChatPCCOE. Because there is such a great reliance on WebSockets in ChatPCCOE we had to ensure we had controls in place to manage subscriptions and prevent memory leaks and performance

problems. Therefore, we implemented cleanup mechanisms that would unsubscribe from any channel that was no longer in use.

We also encountered N+1 query performance degradation when evaluating performance on the feed module as a result of executing multiple database queries. This was accomplished by optimizing the query by using joins.

Finally, we implemented an integer-based regular expression to identify bad words, curse words, and contextual autoplay unusable materials providing limited functionality. In this area, we found many limitations and we will be addressing the need for more robust options in future improvements.

Another technical challenge we have encountered with ChatPCCOE is browser compatibility when utilizing WebSockets.

Another challenge faced by chat PCCOE was content moderation; our regex-based method of identifying profanity and contextually inappropriate material that can become autoplayed did not work consistently. We also have some limitations we need to solve with a future update. In addition, a problem we encountered with chat PCCOE was that web browsers had difficulty communicating via web sockets.

10. CONCLUSION AND FUTURE SCOPE

10.1. CONCLUSIONS

ChatPCCOE has been created to connect students with the College of Engineering, Chinchwad, Pune in one central place. The goal of this new platform is to encourage real-time conversational interaction between students and administration. Utilising a WebSocket based Supabase Realtime Technology, users will be able to communicate via fast, reliable messaging throughout the campus. This platform will include multiple ways for users to connect such as instant messaging, group conversations and sharing posts, while providing easy access to find out important college information, friends, groups, or any other student-related information. This application features an intuitive and simplistic user interface which makes it easy to navigate and communicate to anyone at any time using either their desktop or mobile device. One of the most appealing aspects of the ChatPCCOE platform is the strong security system developed to protect all users from unauthorised access to their personal information, therefore enhancing users' overall experience and trust on the platform. In order to accomplish this enhanced security level, users have only been granted access to their own data using Row-Level Security (RLS). In addition, filtered words and image moderation have been implemented into the platform to help foster an environment where all users can feel comfortable, safe, and respected. The development process of the platform followed a methodology-based approach, whereby all features were designed, tested, and approved prior to implementation. As a whole, ChatPCCOE takes the form of a scalable innovation that will help educational institutions improve communication, collaboration and the sharing of knowledge.

10.2. FUTURE SCOPE

The ChatPCCOE system has been performing well with many features. There are still multiple avenues to improve on this system with future work being more powerful as well as making it easier to use by incorporating additional technologies that increase the usefulness of the system.

Currently, one of the things that can be improved is how the system checks for content. The current way that the system checks for bad words is a simplistic method of searching for bad words; therefore, the current way of checking cannot find words that are used inappropriately nor can the system find hidden words. By utilizing machine learning to search for content will allow the system to better identify content while also ensuring that users remain safe. Thus allowing the ChatPCCOE system to identify "non-obvious" types of content while maintaining user safety.

An additional way to improve the ChatPCCOE system is adding video and/or voice call technology will allow for real-time conversations between users instead of just

sending each other text messages. Users will now be able to conduct meetings, participate in discussions and collaborate more effectively on group projects. Video and/or voice calls will provide great resources for students and teachers to help facilitate academic- related conversations. The ChatPCCOE system will continue to be considered a tool for student/teacher academic interaction as well as a tool for discussion of group projects.

Adding a calendar function would allow users to manage their schedules of specific events like exams, assignments or meeting up with friends. The system would also generate reminders for these appointments so users don’t miss them.

Connecting students to alumni would provide them with professional career advice from people who have already obtained jobs in the field they are interested in since they also went through the same program. Students can benefit from this mentor/mentee relationship to increase their employment opportunities and ultimately their success. Connecting alumni with current students through the ChatPCCOE system is an excellent way to provide the students with advice on their career interests.

Increasing the availability/usage of the ChatPCCOE system by making it mobile will increase overall user experience and increase the likelihood of continued use. Since the ChatPCCOE system is already web-based and accessible via phones and tablets, creating mobile apps could potentially improve performance because they would be faster and more efficient. Some aspects of the ChatPCCOE system could even function offline, making them an even more convenient option for users. Ultimately, a mobile application would make the ChatPCCOE system more user-friendly.

There are additional features that can be developed to gather information on how users are accessing the platform as well as to gather user preferences. The analytics being gathered from this improvement will provide leadership with further options for improving the platform, based on user preferences. Additional features could include options to suggest user connections, such as friends and events. Referring to the

analytics, the ChatPCCOE platform will be a better resource for users when the platform has the additional analytics/ reporting components.

As an enhancement to security the messaging feature may have encrypted communication added for all messages. This would ensure secure communications by making sure the only person who could read the message would have the proper information identified to them. The ChatPCCOE platform could also develop functionality that allows users to access certain components of the platform without an internet connection. If the ChatPCCOE platform develops and implements end-to-end encryption for all messages it will significantly increase the level of security of all communications. Overall, there are multiple improvements available to the ChatPCCOE platform. If there need enhancements and new technologies incorporated into the ChatPCCOE platform it will provide Schools and Students with a tremendous tool. The overall future of the ChatPCCOE platform is promising and will provide for significant enhancements and expansion opportunities.

10.3 APPLICATIONS

ChatPCCOE is a great way to communicate with each other on campus and also much more. The ChatPCCOE platform keeps students, teachers and other college staff connected, to work together and to stay up to date. Such a platform allows the college to provide a more efficient way to communicate as well as to save time for everyone.

The ChatPCCOE platform provides a great way to study with your classmates and to stay informed about your studies. The teachers can use this platform to provide information such as when tests are happening, due dates for assignments and important announcements. This helps ensure that all students have the same information.

The ChatPCCOE platform also provides assistance to students after college. The placement office can use this platform to provide information on jobs, internships and other opportunities. This allows all students to receive the same information fairly.

The ChatPCCOE application assists student groups and clubs in planning events and activities as well as communicating with one another in their respective groups. Notifications sent through the ChatPCCOE application allow members of these groups to effectively collaborate. As a result, the members of these groups have an enjoyable experience being part of a larger community, thereby improving their overall college experience.

The ChatPCCOE will also assist with the development of relationships among the student population. The ability of students to connect through the ChatPCCOE application will promote collaboration and knowledge sharing between members of various departments. The ChatPCCOE application will be particularly useful to students collaborating on class projects, in-class discussions and skill development.

APPENDIX A

AI REPORT

FINAL ChatPCCOE Report finallll.pdf

Pimpri Chinchwad College of Engineering

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AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

APPENDIX B

SIMILARITY INDEX REPORT

FINAL ChatPCCOE Report finallll.pdf

Pimpri Chinchwad College of Engineering

Document Details

Submission ID

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Submission Date

May 3, 2026, 1:43 PM GMT+5:30

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File Name

FINAL ChatPCCOE Report finallll.pdf

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86 Pages

19,361 Words

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iThenticate   Page 1 of 92 - Cover Page   Submission ID trn:oid:::3117:585633015

8% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography

Match Groups

151 Not Cited or Quoted 8%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

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Matches with in-text citation present, but no quotation marks

Top Sources

4% Internet sources
5% Publications
0% Submitted works (Student Papers)

Integrity Flags

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No suspicious text manipulations found.

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A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

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4% Internet sources
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The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

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SANCTION REPORT

PCCOE/COMP(RL)/2025-26/14.

Pimpri Chinchwad Education Trust's

Pimpri Chinchwad College of Engineering

Students Funding Proposal Recommendations

Department: Computer Engineering (Regional Language) **Academic Year:** 2025-26

Semester: I MAIL IN No. 715 Date 18/08/25 File **Date:** 14/07/2025

To,

The Director,

PCCOE, Nigdi, Pune-44

46/RND

Through: Dean R&D

Subject: Recommendation for funding of UG project "PCCOECONNECT" for AY 2025-26

Respected Sir,

BTech Project "PCCOE CONNECT" is reviewed and recommended for financial assistance from the Department of Computer Engineering (Regional Language) . Partial work is already done in TY (Technical Seminar). PCCOE Connect is an essential platform that enables communication, collaboration, and community engagement among students at PCCOE. Partial development is completed in TY (Technical Seminar). As the platform continues to grow and provide more value, it is necessary to secure funding for its ongoing maintenance and optimization. This request outlines the costs required to sustain the platform, covering domain registration, cloud hosting services, and tool subscriptions. These services are essential to ensure the platform remains reliable, secure, and scalable.

Component(INR)	Purpose	Cost	Frequency	Total Yearly Cost
Domain Name (pccoeconnect.com)	Domain registration for the platform's professional identity	₹3,000 (for 3 years)	One-time (3 years)	₹3,000
Supabase Subscription	Managed database, authentication, and backend services	$25 (₹2,148.75*)	Monthly X 12	₹25,785
System deploying, domain connection, feature shipping, maintenance & engagement system	Subscription for deployment, connecting domain, easy feature shipping, maintenance, and engagement features	$25 (₹2,148.75*)	Monthly X 12	₹25,785
			Total	**₹ 54,570**

*Note: The dollar-to-rupee conversion rate is 85.95 as of 14th July 2025, and it may vary based on fluctuations in the exchange rate. This will impact the monthly costs in INR.

Value & Justification

- Platform Adoption: PCCOE Connect will be actively used by 10,000+ students for communication, event participation, polls, and campus updates.
- Growth Potential: The platform has significant growth potential, with plans for additional features like placement boards, alumni networking, and skill-matching algorithms.
- Security & User Engagement: The technologies subscriptions and costs ensure robust data security, real-time data sync, and reliability. The Communication & Engagement System ensures continuous user engagement, notifications, and high retention rates.
- Scalability: These ongoing expenses ensure smooth operations and future-proof the platform by making scaling and feature upgrades seamless.

The summary of the proposals recommended for financial assistance from the department is given below for your kind consideration.

Sr. No.	Project Team and Mentor	Project Title	Budget (Rs.)	Recommended amount
1	Mangal Singhal Anshul Wagh Vedant Sonawane Ms. Rucha Shinde (Asst. Professor)	PCCOE Connect – The Official Student Networking Platform	**₹ 54,570**	**₹ 54,570**
			Total Budget (in Rs.)	**₹ 54,570**

We request approval for a budget of **₹ 54,570** to sustain the platform, covering domain registration, cloud hosting services, and tool subscriptions. This budget will ensure the continued success and growth of PCCOE Connect.

Mrs. Priyanka Gupta
Project Coordinator

Ms. Rucha Shinde
R & D Coordinator

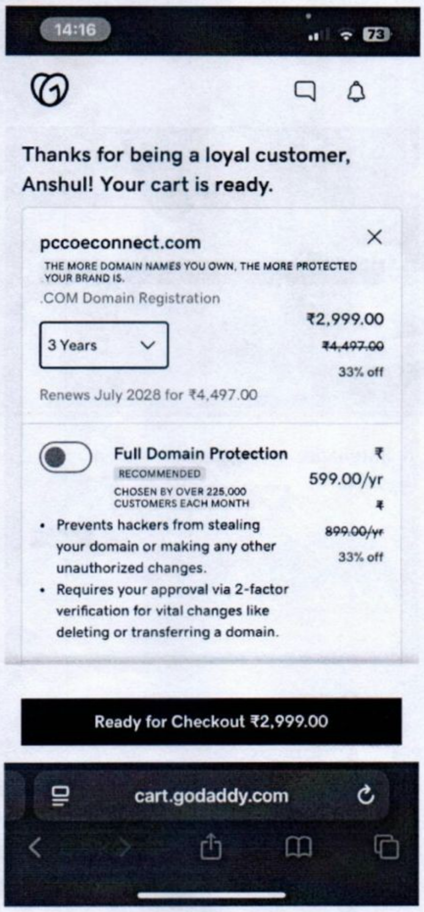
Dr. Rachana Patil
Head of Department

Submitted to Hon. Director sir for information & further approval pl.
18-8-25

Submit to Trust Office for approval

This is only BTT project from RL proposed for funding. funding is proposed for database and deployment. Please check the domainname with PCCOE ~~that~~ website development team.

1) Domain Name (pccoeconnect.com)

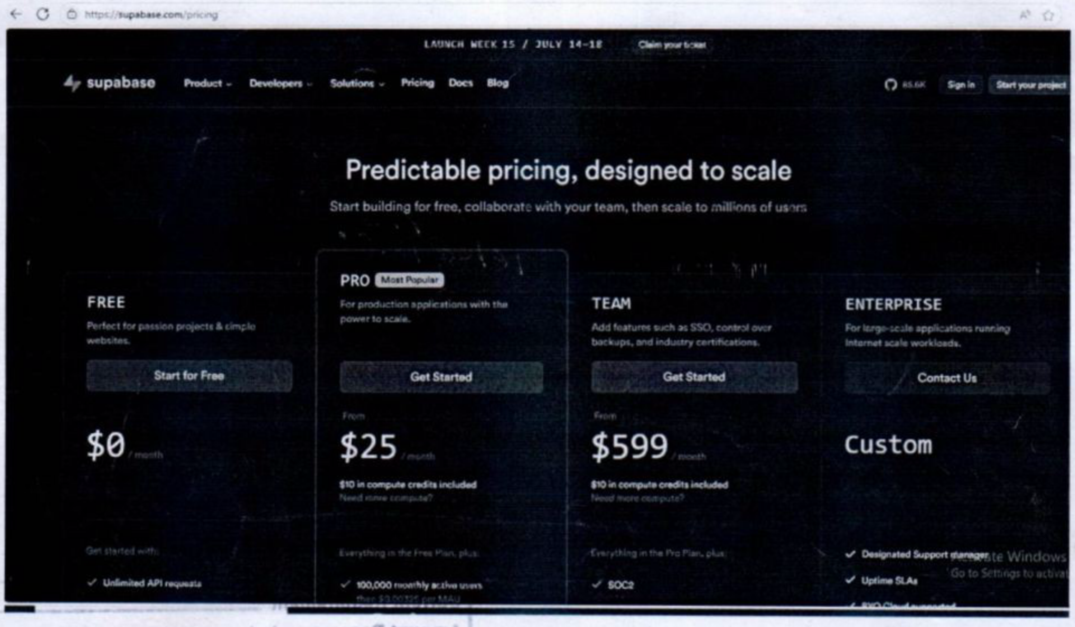


46/RND

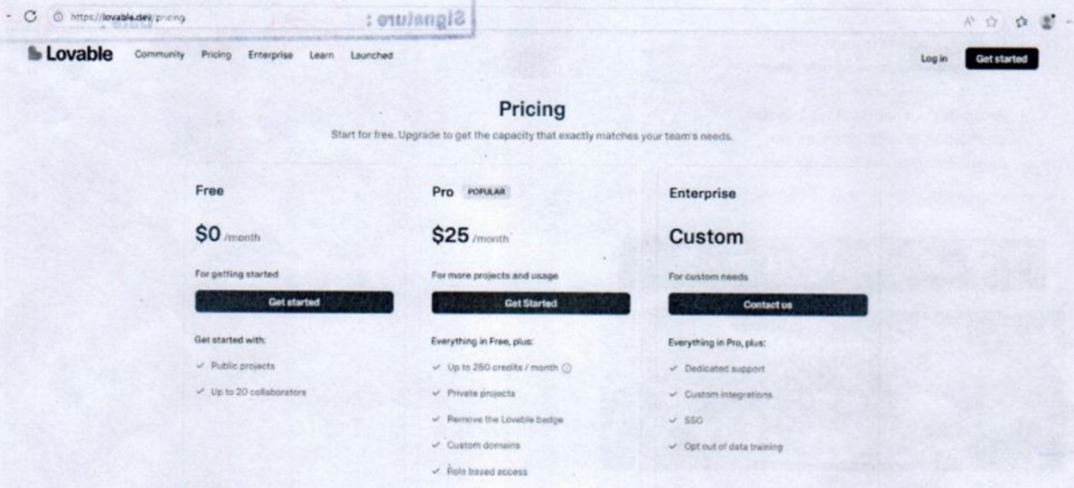
P.C.E.T.'s PIMPRI CHINCHWAD COLLEGE OF ENG...
SECTOR NO. 26, PRADHIKARAN, NIGDI, PUNE - 4...

Dean Office & No.: 46/RND Dept.: COMP-RL
Budget Head RND/4
Allocated Budget 8500000/-
Utilized Budget 5076844/-
Current Requirement 54570/-
Amount Recommended 54570/-
Balance Budget 3368586/-
Signature: Date: 7/8/2025

2) Supabase Subscription



3) System deploying, domain connection, feature shipping, maintenance & engagement system



AI USE DISCLOSURE

We, **Mangal Singhal (PRN:122B1D058)**, **Anshul Wagh (PRN:122B1D066)**, and **Vedant Sonawane (PRN:122B1D060)**, students of Final Year B.Tech, Department of Computer Engineering (Regional Language), hereby disclose that during the preparation of the Major Project Report titled **"ChatPCCOE: A Web-Based Communication and Collaboration Platform for PCCOE Staff and Students"**, we have made use of the following Artificial Intelligence (AI) tools in a controlled, ethical, and assistive manner, as permitted under the PCCOE AI Policy (AY 2025–26, Even Semester).

The AI tool(s) used, along with their purpose and extent of use, are as follows:

1. **Lovable** – used for frontend code generation and UI development.
2. **Claude** – used for research assistance and debugging.
3. **ChatGPT** – used for improving writing, grammar, and content presentation.

The final content we submitted reflects our understanding, analysis, and rewriting.

We further declare that the intellectual content, design decisions, experimental work, analysis, and conclusions presented in this report are entirely our own. The originality of the work has not been compromised through the use of any AI tool. All AI-assisted content has been reviewed, verified, and rewritten by us, and proper citations have been provided wherever applicable. No AI-generated text has been submitted as original work without explicit disclosure.

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